

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

CHAPTER 4

Continuity

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

How to read this book

Everything set in the main column below is Rudin, verbatim. You can read it straight through and never look at an annotation. The annotation layer sits in two places, both outside his text.

Margin notes. A small superscript number in Rudin's text — like this¹ — keys to a numbered note in the outer margin. Since the book prints in black and white, the *tier* of a note is carried by how its number is set:

¹ *The marker is the only mark added to his page, and it is the same for every tier so his line spacing is undisturbed. The number out here tells you what kind of note it is.*

7

Plain number — unpacking. What the sentence is actually saying, where a formula came from, an omitted algebraic step.

7

Outlined number — going deeper. Context beyond the course: what breaks in general, which later theorem this sets up. Skippable.

7

Reversed number — warning. A specific belief students form here that is false, or a place where Rudin's wording will mislead you.

Interleaved boxes. When a note runs longer than a margin will hold, it becomes a box between Rudin's paragraphs, framed by the same three tiers: a **solid** left rule for unpacking, a **dashed** left rule for going deeper, and rules **top and bottom** with a black tab for a warning.

Every figure is an addition. Rudin's Chapter 1 contains no diagrams at all. Any picture you see here is annotation, drawn to expose geometry that his algebra states without illustrating; each carries a grey italic caption.

Where the text is corrected. The main column follows a retyped edition whose own foreword concedes that fresh typos were probably introduced. Where one is evident, the reading is restored from Rudin's third edition and a warning-tier note records the change — silent emendation would make “verbatim” meaningless.

Structural commentary. Two further boxes carry no mathematics at all. They explain the book rather than its contents — why a chapter exists, why Rudin ordered it as he did, why a section that looks like a detour is not one.

HEAVY FRAME, BLACK TITLE BAR

Opens and closes every chapter: what it is for, how it is organized and why in that order, which parts are load-bearing. Where Rudin has explained his own choices — usually in the Preface — these boxes quote him rather than guess.

THIN FRAME, INLINE HEADING | Opens a section: why this material sits here, and what to hold on to.

Books cited in the notes. Where an annotation says another treatment does something differently, it means one of these. They are named because seeing the same theorem set up two ways is often what makes Rudin’s choice legible.

- | | |
|-------------------------------|--|
| <i>Tao, Analysis I–II</i> | Terence Tao, 3rd ed. Builds $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from the Peano axioms and constructs $\mathbb{R}$ by Cauchy sequences rather than cuts. Volume II covers metric spaces and uniform convergence — Rudin’s Chapters 2, 4 and 7. |
| <i>Yu Pin, Tsinghua notes</i> | 数学分析讲义, Tsinghua / Yau Mathematical Sciences Center. Axiomatizes $\mathbb{R}$ by field, order, archimedean and <i>nested interval</i> axioms, then derives the supremum principle. |
| <i>ECNU text</i> | 数学分析 (Mathematical Analysis), 5th ed., 2018, East China Normal University. Standard Chinese two-volume text. Its §7.1 collects six equivalent formulations of completeness and proves them interderivable. |
| <i>USTC notes</i> | 中国科学技术大学少年班数学分析讲义 (程艺 et al., 2005). Teaches the main thread of calculus first and defers real-number theory to a later volume; lists Rudin among its three principal references. |

Chapter 4

CONTINUITY

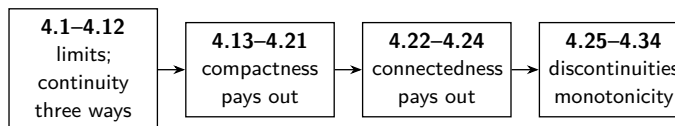
WHY THIS CHAPTER EXISTS

Chapters 2 and 3 built machinery — topology and limits — and this chapter is where it gets spent. Continuity is defined three equivalent ways (ε - δ at 4.5, sequences via 4.2, inverse images at 4.8), and then the two great theorems of Chapter 2 are converted into the two facts every calculus student uses without proof: a continuous function on a compact set is bounded, attains its extrema, and is uniformly continuous (4.14–4.19); a continuous function on an interval takes every intermediate value (4.22–4.23).

The chapter’s centre of gravity is **Theorem 4.8**: continuous means *inverse images of open sets are open*. It converts continuity from a pointwise, quantifier-laden condition into a single structural statement, and it is the form in which every later argument uses the hypothesis.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin’s own account, from the opening paragraph: the theorems “would not become any easier if we restricted ourselves to real functions,” and it “actually simplifies and clarifies the picture to discard unnecessary hypotheses and to state and prove theorems in an appropriately general context.” So everything is done for metric spaces, exactly as in Chapters 2 and 3, and specialised only when the real line genuinely matters (monotonicity, 4.28–4.30, needs an order; the intermediate value theorem needs intervals).



Three things to know in advance.

The proofs are translations, not inventions. Almost every proof here either translates a sequence theorem through the bridge 4.2 (so Chapter 3 does the work), or pulls a topological property of the target space back through f^{-1} (so Chapter 2 does the work). The strategy index of Chapter 1 gains an eighth

pattern: *transport structure along f or f^{-1} .*

The counterexamples are half the content. 4.20–4.21 show exactly what dies without compactness, and 4.24, 4.27 what the intermediate value property does *not* imply. Rudin places a counterexample after nearly every theorem in this chapter; read them as part of the theorem.

Uniform continuity (4.18–4.19) is the new concept.

Everything else repackages Chapters 2–3; uniformity — one δ serving every point at once — is genuinely new, is the hardest distinction in the chapter, and is the hypothesis integration will need in Chapter 6 (Theorem 6.8).

The function concept and some of the related terminology were introduced in Definitions 2.1 and 2.2. Although we shall (in later chapters) be mainly interested in real and complex functions (i.e., in functions whose values are real or complex numbers) we shall also discuss vector-valued functions (i.e., functions with values in $\mathbb{R}^k$) and functions with values in an arbitrary metric space. The theorems we shall discuss in this general setting would not become any easier if we restricted ourselves to real functions, for instance, and it actually simplifies and clarifies the picture to discard unnecessary hypotheses and to state and prove theorems in an appropriately general context.

The domains of definition of our functions will also be metric spaces, suitably specialized in various instances.

LIMITS OF FUNCTIONS

4.1 Definition. Let X and Y be metric spaces; suppose $E \subset X$, f maps E into Y , and p is a limit point of E . We write $f(x) \rightarrow q$ as $x \rightarrow p$, or

$$\lim_{x \rightarrow p} f(x) = q \quad (4.1)$$

if there is a point $q \in Y$ with the following property: For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \quad (4.2)$$

for all points $x \in E$ for which

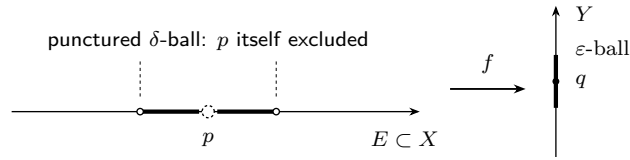
$$0 < d_X(x, p) < \delta. \quad (4.3)$$

The symbols d_X and d_Y refer to the distances in X and Y , respectively.²

If X and/or Y are replaced by the real line, the complex plane, or by some euclidean space $\mathbb{R}^k$, the distances d_X , d_Y are of course replaced by absolute values, or by norms of differences (see Example 2.16).

2 Two structural points hide in (4.3). The strict inequality $0 < d_X(x, p)$ deletes the point p itself: the limit ignores what happens at p , and even whether f is defined there. And p must be a limit point of E — otherwise no x satisfies (4.3) for small δ and every q would qualify vacuously. The hypothesis is exactly what makes the limit unique.

It should be noted that $p \in X$, but that p need not be a point of E in the above definition. Moreover, even if $p \in E$, we may very well have $f(p) \neq \lim_{x \rightarrow p} f(x)$.



The contract of 4.1: however small an ε -ball is drawn about q , some punctured δ -ball about p is mapped entirely into it. The value $f(p)$ — if it exists at all — plays no part.

We can recast this definition in terms of limits of sequences:

4.2 Theorem. Let X, Y, E, f , and p be as in Definition 4.1. Then

$$\lim_{x \rightarrow p} f(x) = q \quad (4.4)$$

if and only if

$$\lim_{n \rightarrow \infty} f(p_n) = q \quad (4.5)$$

for every sequence (p_n) in E such that

$$p_n \neq p, \quad \lim_{n \rightarrow \infty} p_n = p. \quad (4.6)$$

Proof. Suppose (4.4) holds. Choose (p_n) in E satisfying (4.6). Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that $d_Y(f(x), q) < \varepsilon$ if $x \in E$ and $0 < d_X(x, p) < \delta$. Also, there exists N such that $n > N$ implies $0 < d_X(p_n, p) < \delta$. Thus, for $n > N$, we have $d_Y(f(p_n), q) < \varepsilon$, which shows that (4.5) holds.

Conversely, suppose (4.4) is false. Then there exists some $\varepsilon > 0$ such that for every $\delta > 0$ there exists a point $x \in E$ (depending on δ), for which $d_Y(f(x), q) \geq \varepsilon$ but $0 < d_X(x, p) < \delta$. Taking $\delta_n = 1/n$ ($n = 1, 2, 3, \dots$), we thus find a sequence in E satisfying (4.6) for which (4.5) is false.³ $\square$

3 The forward direction chains two contracts: ε buys a δ from f , and that δ buys an N from the sequence. The converse is proved contrapositively, and note its shape: negating “for every ε there is δ ” yields one bad ε whose every δ fails — and harvesting one witness x_n for each $\delta_n = 1/n$ assembles the bad sequence. (Silently, a choice is made for each n ; compare the note at 2.12.)

Corollary. If f has a limit at p , this limit is unique.

This follows from Theorems 3.2(b) and 4.2.

The bridge, and which way to cross it

Theorem 4.2 is a dictionary: function limits on one side, sequence limits on the other. Its value is strategic, and each direction has a distinct use.

To import theorems, cross left to right. All of Chapter 3’s limit arithmetic transfers instantly — that is the entire proof of 4.4, one line long. Anything proved for sequences is proved for function limits, free.

To refute limits, cross right to left. To show $\lim_{x \rightarrow p} f(x)$

does not exist, produce *two* sequences approaching p along which f tends to different values — the function-limit analogue of the two-subsequence trick of 3.1(d). Example 4.27(d) will use exactly this on $\sin(1/x)$ -type behaviour.

The quantifier “*every* sequence” is what makes the dictionary faithful, and it is why the refutation direction is cheap while the verification direction through sequences is usually hopeless — one cannot check all sequences one by one. For verifying, ε - δ or 4.8 are the tools.

4.3 Definition. Suppose we have two complex functions, f and g , both defined on E . By $f + g$ we mean the function which assigns to each point x of E the number $f(x) + g(x)$. Similarly we define the difference $f - g$, the product fg , and the quotient f/g of two functions, with the understanding that the quotient is defined only at those points x of E at which $g(x) \neq 0$. If f assigns to each point x of E the same number c , then f is said to be a *constant* function, or simply a *constant*, and we write $f = c$. If f and g are real functions, and if $f(x) \geq g(x)$ for every $x \in E$, we shall sometimes write $f \geq g$, for brevity.

Similarly, if $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, we define $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ by

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x), \quad (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{f}(x) \cdot \mathbf{g}(x);$$

and if λ is a real number, $(\lambda \mathbf{f})(x) = \lambda \mathbf{f}(x)$.

4.4 Theorem. Suppose $E \subset X$, a metric space, p is a limit point of E , f and g are complex functions on E , and

$$\lim_{x \rightarrow p} f(x) = A, \quad \lim_{x \rightarrow p} g(x) = B.$$

Then

- (a) $\lim_{x \rightarrow p} (f + g)(x) = A + B$;
- (b) $\lim_{x \rightarrow p} (fg)(x) = AB$;
- (c) $\lim_{x \rightarrow p} \left(\frac{f}{g} \right)(x) = \frac{A}{B}$, if $B \neq 0$.

Proof. In view of Theorem 4.2, these assertions follow immediately from the analogous properties of sequences (Theorem 3.3).⁴ $\square$

Remark: If $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, then (a) remains true, and (b) becomes

$$(b') \quad \lim_{x \rightarrow p} (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{A} \cdot \mathbf{B}.$$

(Compare Theorem 3.4.)

4 The dictionary in action: not one ε appears, because every ε was already spent in 3.3. This is the chapter's characteristic proof — one sentence of translation — and the same sentence will prove 4.9. When a proof is this short, the work was done elsewhere; the skill is knowing where.

THREE DEFINITIONS CONVERGING | Continuity arrives here in its ε - δ form (4.5), is linked to limits (4.6) and hence to sequences (through 4.2), and is then recast as the inverse-image condition (4.8). All three are the same property; each has its use. ε - δ verifies concrete examples (4.11); sequences refute; 4.8 proves theorems — it is the only form in which the compactness and connectedness arguments of the next two sections can even be stated. The section's quiet workhorses are 4.7 (compositions) and the assembly line of 4.9–4.11, which manufactures the continuity of every rational function once and for all.

4.5 Definition. Suppose X and Y are metric spaces, $E \subset X$, $p \in E$, and f maps E into Y . Then f is said to be *continuous at p* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \varepsilon$$

for all points $x \in E$ for which $d_X(x, p) < \delta$.

If f is continuous at every point of E , then f is said to be *continuous on E* .

It should be noted that f has to be defined at the point p in order to be continuous at p . (Compare this with the remark following Definition 4.1.)⁵

If p is an isolated point of E , then our definition implies that every function f which has E as its domain of definition is continuous at p . For, no matter which $\varepsilon > 0$ we choose, we can pick $\delta > 0$ so that the only point $x \in E$ for which $d_X(x, p) < \delta$ is $x = p$; then

$$d_Y(f(x), f(p)) = 0 < \varepsilon.$$

6

4.6 Theorem. In the situation given in Definition 4.5, assume also that p is a limit point of E . Then f is continuous at p if and only if $\lim_{x \rightarrow p} f(x) = f(p)$.

Proof. This is clear if we compare Definitions 4.1 and 4.5. □

We now turn to compositions of functions. A brief statement of the following theorem is that a continuous function of a continuous function is continuous.

4.7 Theorem. Suppose X, Y, Z are metric spaces, $E \subset X$, f maps E into Y , g maps the range of f , $f(E)$, into Z , and h is the mapping of E into Z defined by

$$h(x) = g(f(x)) \quad (x \in E).$$

If f is continuous at a point $p \in E$ and if g is continuous at the point $f(p)$, then h is continuous at p .

5 The two definitions differ in exactly two characters: 4.5 requires $p \in E$ where 4.1 required p to be a limit point, and 4.5 drops the $0 <$ from the distance condition — the point p itself is now tested, and the target is $f(p)$, not a free q . Keep the ledger straight: limits ignore p ; continuity is about p .

6 Not a pathology but a consistency check: at an isolated point there is no approach to make, so nothing to fail. It is also why 4.6 needs its extra hypothesis — at an isolated point the limit in 4.6 is undefined (4.1 demands a limit point) while continuity holds vacuously. Every sequence-in- $\mathbb{Z}$, for instance, is a continuous function.

This function h is called the *composition* or the *composite* of f and g . The notation

$$h = g \circ f$$

is frequently used in this context.

Proof. Let $\varepsilon > 0$ be given. Since g is continuous at $f(p)$, there exists $\eta > 0$ such that

$$d_Z(g(y), g(f(p))) < \varepsilon \quad \text{if } d_Y(y, f(p)) < \eta \text{ and } y \in f(E).$$

Since f is continuous at p , there exists $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \eta \quad \text{if } d_X(x, p) < \delta \text{ and } x \in E.$$

It follows that

$$d_Z(h(x), h(p)) = d_Z(g(f(x)), g(f(p))) < \varepsilon$$

if $d_X(x, p) < \delta$ and $x \in E$. Thus h is continuous at p .⁷ □

4.8 Theorem. A mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y .

(Inverse images are defined in Definition 2.2.) This is a very useful characterization of continuity.

Proof. Suppose f is continuous on X and V is an open set in Y . We have to show that every point of $f^{-1}(V)$ is an interior point of $f^{-1}(V)$. So, suppose $p \in X$ and $f(p) \in V$. Since V is open, there exists $\varepsilon > 0$ such that $y \in V$ if $d_Y(f(p), y) < \varepsilon$; and since f is continuous at p , there exists $\delta > 0$ such that $d_Y(f(x), f(p)) < \varepsilon$ if $d_X(x, p) < \delta$. Thus $x \in f^{-1}(V)$ as soon as $d_X(x, p) < \delta$.

Conversely, suppose $f^{-1}(V)$ is open in X for every open set V in Y . Fix $p \in X$ and $\varepsilon > 0$, let V be the set of all $y \in Y$ such that $d_Y(y, f(p)) < \varepsilon$. Then V is open; hence $f^{-1}(V)$ is open; hence there exists $\delta > 0$ such that $x \in f^{-1}(V)$ as soon as $d_X(p, x) < \delta$. But if $x \in f^{-1}(V)$, then $f(x) \in V$, so that $d_Y(f(x), f(p)) < \varepsilon$.

This completes the proof. □

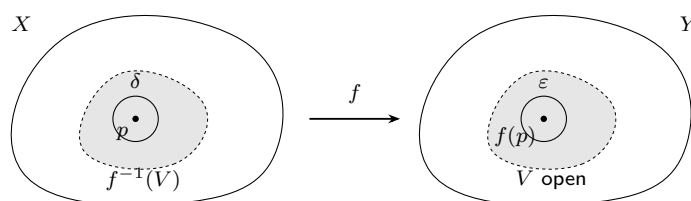
HOW THE PROOF FLOWS

1. **See the two vocabularies.** Continuity speaks in ε - δ ; openness speaks in interior points, i.e. in balls. The theorem claims they say the same thing, so the proof must be a two-way translation, and each direction is found by matching ball to ball.
2. **Forward.** To show $f^{-1}(V)$ open, stand at $p \in f^{-1}(V)$. Openness of V supplies an ε -ball inside V around $f(p)$; continuity at p converts it into a δ -ball around p that f sends into it. That δ -ball certifies p interior. Note the order: V speaks first, conti-

⁷ A relay race, run backwards from the goal: the final tolerance ε is handed to g , which quotes its price η ; that price becomes the tolerance handed to f , which quotes δ . Chains of quantifiers always compose in this order — last function first — and the middle quantity η is bought from one contract and spent on the other. The same relay reappears whenever composed processes must be estimated (e.g. 7.12).

nunity answers.

- 3. Backward.** To verify continuity at p for a given ε , *manufacture* the open set to pull back: the ε -ball V about $f(p)$ — open by 2.19. The hypothesis opens $f^{-1}(V)$, and any δ -ball inside it around p is exactly the δ that continuity demanded.
- 4. Audit.** Each direction uses its hypothesis once, on the ball handed over by the other side; 2.19 (balls are open) is the only outside fact. The theorem is a change of coordinates, not a computation — and note it is stated for f on all of X , using Remark 4.12's licence to forget ambient sets.



Theorem 4.8: whatever ball fits inside V at $f(p)$, continuity supplies a ball at p mapping into it — so $f^{-1}(V)$ is all interior. The dashed regions correspond under f ; the balls are the currency exchanged.

Corollary. A mapping f of a metric space X into a metric space Y is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y .

This follows from the theorem, since a set is closed if and only if its complement is open, and since $f^{-1}(E^c) = [f^{-1}(E)]^c$ for every $E \subset Y$.⁸

We now turn to complex-valued and vector-valued functions, and to functions defined on subsets of $\mathbb{R}^k$.

4.9 Theorem. Let f and g be complex continuous functions on a metric space X . Then $f + g$, fg , and f/g are continuous on X .

In the last case, we must of course assume that $g(x) \neq 0$, for all $x \in X$.

Proof. At isolated points of X there is nothing to prove. At limit points, the statement follows from Theorems 4.4 and 4.6. $\square$

4.10 Theorem.

- (a) Let $f_1, \dots, f_k$ be real functions on a metric space X , and let $\mathbf{f}$ be the mapping of X into $\mathbb{R}^k$ defined by

$$\mathbf{f}(x) = (f_1(x), \dots, f_k(x)) \quad (x \in X); \quad (4.7)$$

then $\mathbf{f}$ is continuous if and only if each of the functions $f_1, \dots, f_k$ is continuous.

- (b) If $\mathbf{f}$ and $\mathbf{g}$ are continuous mappings of X into $\mathbb{R}^k$, then $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ are continuous on X .

⁸ The complement identity is exactly the third displayed law in the box at 2.2 — inverse images respect complements, images do not. Here is the promised payoff: no image-based version of 4.8 exists, and the asymmetry noted in Chapter 2 becomes the reason this whole chapter runs on f^{-1} .

The functions $f_1, \dots, f_k$ are called the *components* of $\mathbf{f}$. Note that $\mathbf{f} + \mathbf{g}$ is a mapping into $\mathbb{R}^k$, whereas $\mathbf{f} \cdot \mathbf{g}$ is a real function on X .

Proof. Part (a) follows from the inequalities

$$|f_j(x) - f_j(y)| \leq |\mathbf{f}(x) - \mathbf{f}(y)| = \left\{ \sum_{i=1}^k |f_i(x) - f_i(y)|^2 \right\}^{1/2},$$

for $j = 1, \dots, k$. Part (b) follows from (a) and Theorem 4.9.⁹ $\square$

4.11 Examples. If $x_1, \dots, x_k$ are the coordinates of the point $\mathbf{x} \in \mathbb{R}^k$, the functions ϕ_i defined by

$$\phi_i(\mathbf{x}) = x_i \quad (\mathbf{x} \in \mathbb{R}^k) \quad (4.8)$$

are continuous on $\mathbb{R}^k$, since the inequality

$$|\phi_i(\mathbf{x}) - \phi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$$

shows that we may take $\delta = \varepsilon$ in Definition 4.5.

The functions ϕ_i are sometimes called the *coordinate functions*.

Repeated application of Theorem 4.9 then shows that every *monomial*

$$x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} \quad (4.9)$$

where $n_1, \dots, n_k$ are nonnegative integers, is continuous on $\mathbb{R}^k$. The same is true of constant multiples of (4.9), since constants are evidently continuous. It follows that every *polynomial* P , given by

$$P(\mathbf{x}) = \sum c_{n_1 \dots n_k} x_1^{n_1} \cdots x_k^{n_k} \quad (\mathbf{x} \in \mathbb{R}^k), \quad (4.10)$$

is continuous on $\mathbb{R}^k$. Here the coefficients $c_{n_1 \dots n_k}$ are complex numbers, $n_1, \dots, n_k$ are nonnegative integers, and the sum in (4.10) has finitely many terms.

Furthermore, every *rational function* in $x_1, \dots, x_k$, that is, every quotient of two polynomials of the form (4.10), is continuous on $\mathbb{R}^k$ wherever the denominator is different from zero.¹⁰

From the triangle inequality one sees easily that

$$||\mathbf{x}| - |\mathbf{y}|| \leq |\mathbf{x} - \mathbf{y}| \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^k). \quad (4.11)$$

Hence the mapping $\mathbf{x} \rightarrow |\mathbf{x}|$ is a continuous real function on $\mathbb{R}^k$.

If now $\mathbf{f}$ is a continuous mapping from a metric space X into $\mathbb{R}^k$, and if ϕ is defined on X by setting $\phi(p) = |\mathbf{f}(p)|$, it follows, by Theorem 4.7, that ϕ is a continuous real function on X .

4.12 Remark. We defined the notion of continuity for functions defined on a subset E of a metric space X . However, the complement of E in X plays no role whatever in this definition (note that the situation was somewhat different for limits of functions). Accordingly, we lose nothing

9 The same two-sided squeeze as 3.4(a): each component error is at most the vector error, and the vector error is at most $\sqrt{k}$ times the worst component. Coordinatewise reduction, third appearance — it will carry Chapter 9's entire differential calculus.

10 An assembly line, and the economical way to prove continuity ever after: two atoms (constants and coordinates, the latter with the cleanest possible modulus, $\delta = \varepsilon$) and three combining rules (sum, product, quotient from 4.9; composition from 4.7). Almost no continuity proof after this page touches ε again — functions are built out of continuous parts. Chapter 8 extends the reach to power series, where the sums become infinite and uniformity must be paid for.

of interest by discarding the complement of the domain of f . This means that we may just as well talk only about continuous mappings of one metric space into another, rather than of mappings of subsets. This simplifies statements and proofs of some theorems. We have already made use of this principle in Theorems 4.8 to 4.10, and will continue to do so in the following section on compactness.¹¹

[11] This is 2.16's observation — every subset is a metric space — earning its keep, and it is subtler than it looks. Continuity of $f|_E$ is measured inside E : the function $f(x) = 1$ for rational x , 0 otherwise, is wildly discontinuous on $\mathbb{R}$, yet its restriction to $\mathbb{Q}$ is constant, hence continuous. “Continuous on E ” in Rudin always means continuous as a function on the metric space E — the same relativity as “open in E ” (2.29–2.30), managed the same way: name the space.

CONTINUITY AND COMPACTNESS

CHAPTER 2'S INVESTMENT MATURES | Five theorems, one engine. 4.14 shows continuous images of compact sets are compact — a one-paragraph proof, because 4.8 lets the compactness of the domain be spent directly on a pulled-back cover. Everything else in the section is 4.14 plus a Chapter 2 fact: boundedness (4.15, via Heine–Borel), attainment of extrema (4.16, via 2.28), continuity of inverse maps (4.17), and uniform continuity (4.19, where compactness converts infinitely many local δ 's into one global one — the finite-minimum move yet again, and Rudin says so in the proof).

Then the audit: 4.20–4.21 dismantle each theorem when compactness is removed, one counterexample per conclusion. The section is a matched pair — what compactness buys, and the invoice showing nothing was free.

4.13 Definition. A mapping $\mathbf{f}$ of a set E into $\mathbb{R}^k$ is said to be *bounded* if there is a real number M such that $|\mathbf{f}(x)| \leq M$ for all $x \in E$.

4.14 Theorem. Suppose f is a continuous mapping of a compact metric space X into a metric space Y . Then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ be an open cover of $f(X)$. Since f is continuous, Theorem 4.8 shows that each of the sets $f^{-1}(V_\alpha)$ is open. Since X is compact, there are finitely many indices, say $\alpha_1, \dots, \alpha_n$, such that

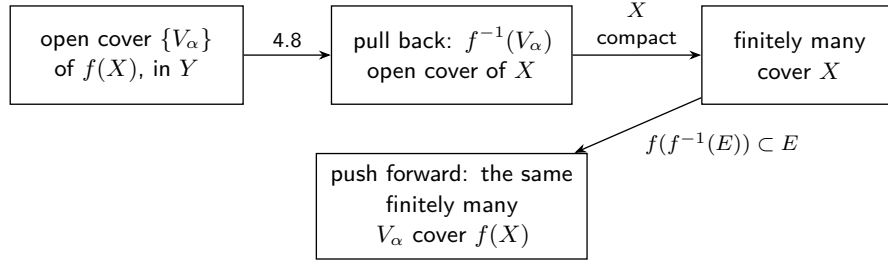
$$X \subset f^{-1}(V_{\alpha_1}) \cup \dots \cup f^{-1}(V_{\alpha_n}). \quad (4.12)$$

Since $f(f^{-1}(E)) \subset E$ for every $E \subset Y$, (4.12) implies that

$$f(X) \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}. \quad (4.13)$$

This completes the proof. □

Note: We have used the relation $f(f^{-1}(E)) \subset E$, valid for $E \subset Y$. If $E \subset X$, then $f^{-1}(f(E)) \supset E$; equality need not hold in either case.



Theorem 4.14 as a flow diagram: pull the cover back, spend compactness, push the finite subcover forward. Each arrow is one theorem, and the round trip works because f is defined on all of X — the push-forward inclusion needs nothing else.

We shall now deduce some consequences of Theorem 4.14.

4.15 Theorem. *If f is a continuous mapping of a compact metric space X into $\mathbb{R}^k$, then $f(X)$ is closed and bounded. Thus, f is bounded.*

This follows from Theorem 2.41. The result is particularly important when f is real:

4.16 Theorem. *Suppose f is a continuous real function on a compact metric space X , and*

$$M = \sup_{p \in X} f(p), \quad m = \inf_{p \in X} f(p). \quad (4.14)$$

Then there exist points $p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

The notation in (4.14) means that M is the least upper bound of the set of all numbers $f(p)$, where p ranges over X , and that m is the greatest lower bound of this set of numbers.

The conclusion may also be stated as follows: *There exist points p and q in X such that $f(q) \leq f(x) \leq f(p)$ for all $x \in X$; that is, f attains its maximum (at p) and its minimum (at q).*

Proof. By Theorem 4.15, $f(X)$ is a closed and bounded set of real numbers; hence $f(X)$ contains

$$M = \sup f(X) \quad \text{and} \quad m = \inf f(X),$$

by Theorem 2.28.¹² □

4.17 Theorem. *Suppose f is a continuous 1-1 mapping of a compact metric space X onto a metric space Y . Then the inverse mapping f^{-1} defined on Y by*

$$f^{-1}(f(x)) = x \quad (x \in X)$$

is a continuous mapping of Y onto X .

Proof. Applying Theorem 4.8 to f^{-1} in place of f , we see that it suffices to prove that $f(V)$ is an open set in Y for every open set V in X . Fix such a set V .

The complement V^c of V is closed in X , hence compact (Theorem 2.35); hence $f(V^c)$ is a compact subset of Y (Theorem 4.14) and so is

12 The extreme value theorem of calculus, in three citations: 4.14 makes the image compact, 2.41 reads that as closed and bounded, 2.28 puts the supremum of a closed bounded set inside it. Every hard step was prepaid in Chapter 2 — exactly as the note at 2.28 promised. Notice the theorem gives no way to find p and q ; pure existence, from compactness alone.

13 A ricochet proof: to open $f(V)$, close its complement. The chain closed $\rightarrow$ compact $\rightarrow$ compact image $\rightarrow$ closed converts domain-compactness into the openness 4.8 demands of f^{-1} — and each link is a Chapter 2 theorem (2.35, 4.14, 2.34). Bijectivity enters exactly once, to say image-of-complement is complement-of-image. Example 4.21 shows the whole chain snaps without compactness.

closed in Y (Theorem 2.34). Since f is one-to-one and onto, $f(V)$ is the complement of $f(V^c)$. Hence $f(V)$ is open.¹³ $\square$

4.18 Definition. Let f be a mapping of a metric space X into a metric space Y . We say that f is *uniformly continuous* on X if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

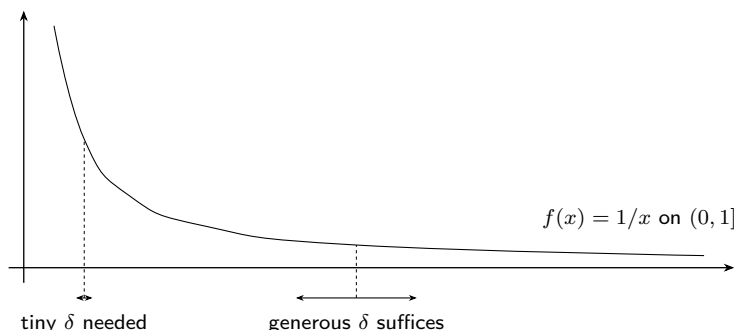
$$d_Y(f(p), f(q)) < \varepsilon \quad (4.15)$$

for all p and q in X for which $d_X(p, q) < \delta$.

Let us consider the differences between the concepts of continuity and of uniform continuity. First, uniform continuity is a property of a function on a set, whereas continuity can be defined at a single point. To ask whether a given function is uniformly continuous at a certain point is meaningless. Second, if f is continuous on X , then it is possible to find, for each $\varepsilon > 0$ and *for each point p of X* , a number $\delta > 0$ having the property specified in Definition 4.5. This δ depends on ε *and on p* . If f is, however, uniformly continuous on X , then it is possible, for each $\varepsilon > 0$, to find *one* number $\delta > 0$ which will do for *all* points p of X .¹⁴

Evidently, every uniformly continuous function is continuous. That the two concepts are equivalent on compact sets follows from the next theorem.

14 Quantifier order is the entire difference: continuity is $\forall p \forall \varepsilon \exists \delta$, uniform continuity is $\forall \varepsilon \exists \delta \forall p$. Moving $\forall p$ past $\exists \delta$ strips δ of its dependence on p . This is the book's sharpest lesson in reading quantifiers, and the figure below is the picture to attach to it.



Continuity that is not uniform: the same ε of vertical tolerance costs ever-smaller δ as the curve steepens toward the missing endpoint. No single δ serves every point — Theorem 4.20(c) makes this the general phenomenon on bounded non-compact sets.

4.19 Theorem. Let f be a continuous mapping of a compact metric space X into a metric space Y . Then f is uniformly continuous on X .

Proof. Let $\varepsilon > 0$ be given. Since f is continuous, we can associate to each point $p \in X$ a positive number $\phi(p)$ such that

$$q \in X, d_X(p, q) < \phi(p) \quad \text{implies} \quad d_Y(f(p), f(q)) < \frac{\varepsilon}{2}. \quad (4.16)$$

Let $J(p)$ be the set of all $q \in X$ for which

$$d_X(p, q) < \frac{1}{2}\phi(p). \quad (4.17)$$

Since $p \in J(p)$, the collection of all sets $J(p)$ is an open cover of X ; since X is compact, there is a finite set of points $p_1, \dots, p_n$ in X , such that

$$X \subset J(p_1) \cup \dots \cup J(p_n). \quad (4.18)$$

We put

$$\delta = \frac{1}{2} \min\{\phi(p_1), \dots, \phi(p_n)\}. \quad (4.19)$$

Then $\delta > 0$. (This is one point where the finiteness of the covering, inherent in the definition of compactness, is essential. The minimum of a finite set of positive numbers is positive, whereas the inf of an infinite set of positive numbers may very well be 0.)

Now let q and p be points of X such that $d_X(p, q) < \delta$. By (4.18), there is an integer m , $1 \leq m \leq n$, such that $p \in J(p_m)$; hence

$$d_X(p, p_m) < \frac{1}{2}\phi(p_m), \quad (4.20)$$

and we also have

$$d_X(q, p_m) \leq d_X(p, q) + d_X(p, p_m) < \delta + \frac{1}{2}\phi(p_m) \leq \phi(p_m).$$

Finally, (4.16) shows that therefore

$$d_Y(f(p), f(q)) \leq d_Y(f(p), f(p_m)) + d_Y(f(q), f(p_m)) < \varepsilon.$$

This completes the proof. $\square$

An alternative proof is sketched in Exercise 4.10.

HOW THE PROOF FLOWS

- 1. Name the enemy.** Continuity hands out a private δ -radius $\phi(p)$ at each point; uniformity demands one public δ . The natural attempt — take $\inf_p \phi(p)$ — fails because an infimum of infinitely many positive numbers can be 0 (Rudin says exactly this in the parenthesis at (4.19)). The proof exists to make that infimum finite.
- 2. Turn the private radii into a cover.** Each point owns the ball $J(p)$ of *half* its radius; together they cover X . Compactness reduces to finitely many centers $p_1, \dots, p_n$ — and now the minimum in (4.19) is over a finite set, hence positive. This is the local-to-global conversion promised in Chapter 2's signpost, performed literally.
- 3. Why half?** The safety margin is what makes the public δ work for pairs. A point p lands in some $J(p_m)$, within $\frac{1}{2}\phi(p_m)$ of the center; its partner q is within $\delta \leq \frac{1}{2}\phi(p_m)$ more, so *both* lie inside the full radius $\phi(p_m)$. Without the halving, q could fall just outside.
- 4. Close through the center.** Neither p nor q is compared with

the other directly — both are compared with p_m , each within $\varepsilon/2$, and the triangle inequality adds the halves. That is why (4.16) was set up with $\varepsilon/2$: working backwards, pattern 4.

- 5. Audit.** Compactness spent once (at the finite subcover), continuity spent once per point (to define ϕ), triangle inequality twice. The halving trick — shrink each local radius so that overlapping neighbours still see each other — recurs in Chapter 7 and in every partition-of-unity argument (Chapter 10).

We now proceed to show that compactness is essential in the hypotheses of Theorems 4.14, 4.15, 4.16, and 4.19.

4.20 Theorem. *Let E be a noncompact set in $\mathbb{R}^1$. Then*

- (a) *there exists a continuous function on E which is not bounded;*
- (b) *there exists a continuous and bounded function on E which has no maximum.*

If, in addition, E is bounded, then

- (c) *there exists a continuous function on E which is not uniformly continuous.*

Proof. Suppose first that E is bounded, so that there exists a limit point x_0 of E which is not a point of E .¹⁵ Consider

$$f(x) = \frac{1}{x - x_0} \quad (x \in E). \quad (4.21)$$

This is continuous on E (Theorem 4.9), but evidently unbounded. To see that (4.21) is not uniformly continuous, let $\varepsilon > 0$ and $\delta > 0$ be arbitrary, and choose a point $x \in E$ such that $|x - x_0| < \delta$. Taking t close enough to x_0 , we can then make the difference $|f(t) - f(x)|$ greater than ε , although $|t - x| < \delta$. Since this is true for every $\delta > 0$, f is not uniformly continuous on E .

The function g given by

$$g(x) = \frac{1}{1 + (x - x_0)^2} \quad (x \in E) \quad (4.22)$$

is continuous on E and is bounded, since $0 < g(x) < 1$. It is clear that

$$\sup_{x \in E} g(x) = 1,$$

whereas $g(x) < 1$ for all $x \in E$. Thus g has no maximum on E .

Having proved the theorem for bounded sets E , let us now suppose that E is unbounded. Then $f(x) = x$ establishes (a), whereas

$$h(x) = \frac{x^2}{1 + x^2} \quad (x \in E) \quad (4.23)$$

¹⁵ Here 2.41 is being run in reverse: noncompact-but-bounded forces not-closed, which hands the proof its weapon — a missing limit point x_0 . Every counterexample in the bounded case is built by blowing up at x_0 ; the theorem is really about how continuity on E cannot see the hole just outside E .

establishes (b), since

$$\sup_{x \in E} h(x) = 1$$

and $h(x) < 1$ for all $x \in E$.

Assertion (c) would be false if boundedness were omitted from the hypotheses. For, let E be the set of all integers. Then every function defined on E is uniformly continuous on E . To see this, we need merely take $\delta < 1$ in Definition 4.18.¹⁶ $\square$

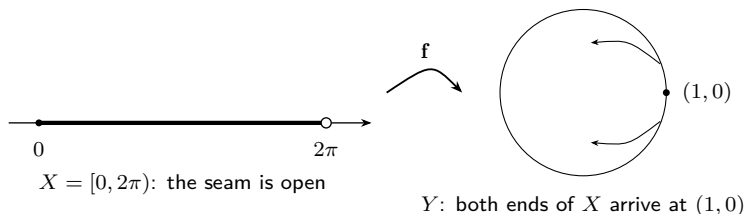
We conclude this section by showing that compactness is also essential in Theorem 4.17.

4.21 Example. Let X be the half-open interval $[0, 2\pi)$ on the real line, and let $\mathbf{f}$ be the mapping of X onto the circle Y consisting of all points whose distance from the origin is 1, given by

$$\mathbf{f}(t) = (\cos t, \sin t) \quad (0 \leq t < 2\pi). \quad (4.24)$$

The continuity of the trigonometric functions cosine and sine, as well as their periodicity properties, will be established in Chap. 8. These results show that $\mathbf{f}$ is a continuous 1-1 mapping of X onto Y .

However, the inverse mapping (which exists, since $\mathbf{f}$ is one-to-one and onto) fails to be continuous at the point $(1, 0) = \mathbf{f}(0)$. Of course, X is not compact in this example. (It may be of interest to observe that $\mathbf{f}^{-1}$ fails to be continuous in spite of the fact that Y is compact!)



Example 4.21: wrapping the half-open interval onto the circle glues the missing endpoint to 0. Points just below 2π land next to $(1, 0)$, but $\mathbf{f}^{-1}$ throws them back near 2π — a jump of almost 2π across the seam. Continuity of the inverse is a property the domain's compactness must fund; the target's cannot.

CONTINUITY AND CONNECTEDNESS

ONE THEOREM, ONE COROLLARY, ONE WARNING | The connectedness investment of 2.45–2.47 pays out in a single exchange: continuous images of connected sets are connected (4.22), and since the connected subsets of $\mathbb{R}$ are exactly the intervals (2.47), the intermediate value theorem falls out in three citations (4.23). The warning is 4.24: the intermediate value *property* does not characterise continuity — $\sin(1/x)$ has it while being badly discontinuous. The converse a beginner expects is false.

[16] So the converse pattern is not clean: noncompactness kills boundedness and attainment outright, but uniform continuity survives on $\mathbb{Z}$ — an unbounded, closed, noncompact set where every function is uniformly continuous because points cannot get close. The moral: (a) and (b) fail through escape (to a missing point or to infinity), (c) fails only through crowding against a missing point. Different failure modes, different hypotheses. Exercise 4.8 probes the same boundary.

4.22 Theorem. *If f is a continuous mapping of a metric space X into a metric space Y , and if E is a connected subset of X , then $f(E)$ is connected.*

Proof. Assume, on the contrary, that $f(E) = A \cup B$, where A and B are nonempty separated subsets of Y . Put $G = E \cap f^{-1}(A)$, $H = E \cap f^{-1}(B)$.

Then $E = G \cup H$, and neither G nor H is empty.

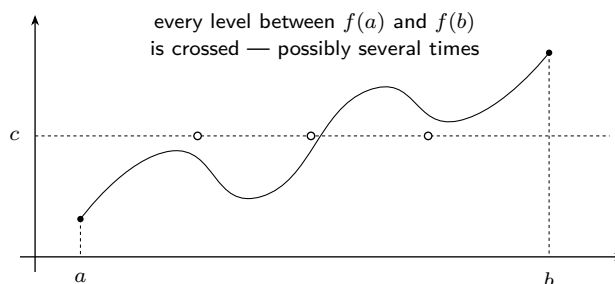
Since $A \subset \bar{A}$ (the closure of A), we have $G \subset f^{-1}(\bar{A})$; the latter set is closed, since f is continuous; hence $\bar{G} \subset f^{-1}(\bar{A})$. It follows that $f(\bar{G}) \subset \bar{A}$. Since $f(H) = B$ and $\bar{A} \cap B$ is empty, we conclude that $\bar{G} \cap H$ is empty.

The same argument shows that $G \cap \bar{H}$ is empty. Thus G and H are separated. This is impossible if E is connected.¹⁷ $\square$

4.23 Theorem. *Let f be a continuous real function on the interval $[a, b]$. If $f(a) < f(b)$ and if c is a number such that $f(a) < c < f(b)$, then there exists a point $x \in (a, b)$ such that $f(x) = c$.*

A similar result holds, of course, if $f(a) > f(b)$. Roughly speaking, the theorem says that a continuous real function assumes all intermediate values on an interval.

Proof. By Theorem 2.47, $[a, b]$ is connected; hence Theorem 4.22 shows that $f([a, b])$ is a connected subset of $\mathbb{R}^1$, and the assertion follows if we appeal once more to Theorem 2.47.¹⁸ $\square$



The intermediate value theorem: a connected image cannot skip the level c . The theorem asserts at least one crossing and says nothing about how many, nor where.

4.24 Remark. At first glance, it might seem that Theorem 4.23 has a converse. That is, one might think that if for any two points $x_1 < x_2$ and for any number c between $f(x_1)$ and $f(x_2)$ there is a point x in (x_1, x_2) such that $f(x) = c$, then f must be continuous.

That this is not so may be concluded from Example 4.27(d).

DISCONTINUITIES

A TAXONOMY, THEN A ZOO | Failure of continuity comes in two grades. If both one-sided limits exist, the failure is a bookkeeping

17 The same transport strategy as 4.14, run contrapositively: a separation of the image pulls back to a separation of the domain. The only technical point is that “separated” involves closures, and this is where the corollary to 4.8 earns its place — $f^{-1}(\bar{A})$ is closed, so it swallows $\bar{G}$, keeping the pulled-back pieces apart. Images transport connectedness and compactness; inverse images transport openness and closedness. Keep the direction ledger straight.

18 Three citations and no computation: 2.47 converts “interval” to “connected,” 4.22 pushes connectedness through f , and 2.47 converts back to “interval” — which by definition contains everything between $f(a)$ and $f(b)$. Chapter 1’s Theorem 1.21 was this argument done by hand, before any of the three tools existed; compare the box there. Exercise 4.14 (fixed points) is the standard application.

error — a jump, or a wrong value — and is called *simple* (first kind). If a one-sided limit fails to exist, the function oscillates or scatters, and the discontinuity is of the second kind. Example 4.27 exhibits all the species; (a) and (b) are the standard pathologies built from $\mathbb{Q}$, and (d) is the oscillator that killed the converse of the intermediate value theorem in 4.24. The taxonomy exists for the next section: monotonic functions will be shown to admit only the first kind, and only countably many.

If x is a point in the domain of definition of the function f at which f is not continuous, we say that f is *discontinuous at x* , or that f has a *discontinuity at x* . If f is defined on an interval or on a segment, it is customary to divide discontinuities into two types. Before giving this classification, we have to define the *right-hand* and the *left-hand limits* of f at x , which we denote by $f(x+)$ and $f(x-)$, respectively.

4.25 Definition. Let f be defined on (a, b) . Consider any point x such that $a \leq x < b$. We write

$$f(x+) = q$$

if $f(t_n) \rightarrow q$ as $n \rightarrow \infty$, for all sequences (t_n) in (x, b) such that $t_n \rightarrow x$. To obtain the definition of $f(x-)$, for $a < x \leq b$, we restrict ourselves to sequences (t_n) in (a, x) .

It is clear that for any point x of (a, b) , $\lim_{t \rightarrow x} f(t)$ exists if and only if

$$f(x+) = f(x-) = \lim_{t \rightarrow x} f(t).$$

4.26 Definition. Let f be defined on (a, b) . If f is discontinuous at a point x , and if $f(x+)$ and $f(x-)$ exist, then f is said to have a *discontinuity of the first kind*, or a *simple discontinuity*, at x . Otherwise the discontinuity is said to be *of the second kind*.

There are two ways in which a function can have a simple discontinuity: either $f(x+) \neq f(x-)$ [in which case the value $f(x)$ is immaterial], or $f(x+) = f(x-) \neq f(x)$.

4.27 Examples.

(a) Define

$$f(x) = \begin{cases} 1 & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

Then f has a discontinuity of the second kind at every point x , since neither $f(x+)$ nor $f(x-)$ exists.¹⁹

(b) Define

$$f(x) = \begin{cases} x & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

¹⁹ The Dirichlet function, and the standard witness that “discontinuous everywhere” is possible. Both $\mathbb{Q}$ and its complement are dense (1.20(b) and Exercise 1.1), so every one-sided approach can be made along either set, producing both 0 and 1 as subsequential limits — the refutation direction of the bridge 4.2. It returns in Chapter 6 as the function that is not Riemann-integrable (Exercise 6.4).

20 Continuity at exactly one point — worth a pause, since beginners often believe continuity somewhere forces decency nearby. The squeeze $|f(x)| \leq |x|$ gives continuity at 0 no matter how wild the values elsewhere.

Then f is continuous at $x = 0$ and has a discontinuity of the second kind at every other point.²⁰

(c) Define

$$f(x) = \begin{cases} x + 2 & (-3 < x < -2), \\ -x - 2 & (-2 \leq x < 0), \\ x + 2 & (0 \leq x < 1). \end{cases}$$

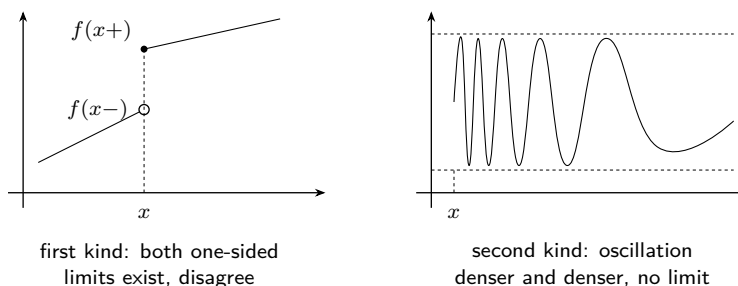
Then f has a simple discontinuity at $x = 0$ and is continuous at every other point of $(-3, 1)$.

(d) Define

$$f(x) = \begin{cases} \sin \frac{1}{x} & (x \neq 0), \\ 0 & (x = 0). \end{cases}$$

Since neither $f(0+)$ nor $f(0-)$ exists, f has a discontinuity of the second kind at $x = 0$. We have not yet shown that $\sin x$ is a continuous function. If we assume this result for the moment, Theorem 4.7 implies that f is continuous at every point $x \neq 0$.²¹

21 The oscillator that closes 4.24: on any interval $(0, \delta)$ it sweeps all of $[-1, 1]$, so it has the full intermediate value property, yet it is discontinuous at 0. The intermediate value property constrains what a function skips, not how it settles — and settling is what continuity is. Exercise 4.14's fixed-point result survives for such functions; their integrability will be delicate in Chapter 6.



The two kinds. Left, a jump (4.27(c)): the function arrives at different platforms from the two sides. Right, $\sin(1/x)$ near 0 (4.27(d)): the one-sided limits do not exist at all, because every neighbourhood of x sweeps the whole band.

MONOTONIC FUNCTIONS

We shall now study those functions which never decrease (or never increase) on a given segment.

4.28 Definition. Let f be real on (a, b) . Then f is said to be *monotonically increasing* on (a, b) if $a < x < y < b$ implies $f(x) \leq f(y)$. If the last inequality is reversed, we obtain the definition of a *monotonically decreasing* function. The class of *monotonic* functions consists of both the increasing and the decreasing functions.

4.29 Theorem. Let f be monotonically increasing on (a, b) . Then $f(x+)$ and $f(x-)$ exist at every point x of (a, b) . More precisely,

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t). \quad (4.25)$$

Furthermore, if $a < x < y < b$, then

$$f(x+) \leq f(y-). \quad (4.26)$$

Analogous results evidently hold for monotonically decreasing functions.

Proof. By hypothesis, the set of numbers $f(t)$, where $a < t < x$, is bounded above by the number $f(x)$, and therefore has a least upper bound which we shall denote by A . Evidently $A \leq f(x)$. We have to show that $A = f(x-)$.

Let $\varepsilon > 0$ be given. It follows from the definition of A as a least upper bound that there exists $\delta > 0$ such that $a < x - \delta < x$ and

$$A - \varepsilon < f(x - \delta) \leq A. \quad (4.27)$$

Since f is monotonic, we have

$$f(x - \delta) \leq f(t) \leq A \quad (x - \delta < t < x). \quad (4.28)$$

Combining (4.27) and (4.28), we see that

$$|f(t) - A| < \varepsilon \quad (x - \delta < t < x).$$

Hence $f(x-) = A$.²²

The second half of (4.25) is proved in precisely the same way.

Next, if $a < x < y < b$, we see from (4.25) that

$$f(x+) = \inf_{x < t < b} f(t) = \inf_{x < t < y} f(t). \quad (4.29)$$

The last equality is obtained by applying (4.25) to (a, y) in place of (a, b) . Similarly,

$$f(y-) = \sup_{a < t < y} f(t) = \sup_{x < t < y} f(t). \quad (4.30)$$

Comparison of (4.29) and (4.30) gives (4.26). $\square$

Corollary. *Monotonic functions have no discontinuities of the second kind.*

This corollary implies that every monotonic function is discontinuous at a countable set of points at most. Instead of appealing to the general theorem whose proof is sketched in Exercise 4.17, we give here a simple proof which is applicable to monotonic functions.

4.30 Theorem. *Let f be monotonic on (a, b) . Then the set of points of (a, b) at which f is discontinuous is at most countable.*

Proof. Suppose, for the sake of definiteness, that f is increasing, and let E be the set of points at which f is discontinuous.

With every point x of E we associate a rational number $r(x)$ such that

$$f(x-) < r(x) < f(x+).$$

²² The bounded-implies-sup pattern of the strategy index (pattern 3), in its function-limit costume: monotonicity makes the left values bounded, the supremum A exists by 1.19, and clause (ii) of 1.8 plants a term within ε ; monotonicity then drags the whole approach into the gap — exactly the mechanism of 3.14, with $t \uparrow x$ in place of $n \rightarrow \infty$.

Since $x_1 < x_2$ implies $f(x_1+) \leq f(x_2-)$, we see that $r(x_1) \neq r(x_2)$ if $x_1 \neq x_2$.

We have thus established a 1-1 correspondence between the set E and a subset of the set of rational numbers. The latter, as we know, is countable.²³ $\square$

4.31 Remark. It should be noted that the discontinuities of a monotonic function need not be isolated. In fact, given any countable subset E of (a, b) , which may even be dense, we can construct a function f , monotonic on (a, b) , discontinuous at every point of E , and at no other point of (a, b) .

To show this, let the points of E be arranged in a sequence (x_n) , $n = 1, 2, 3, \dots$. Let (c_n) be a sequence of positive numbers such that $\sum c_n$ converges. Define

$$f(x) = \sum_{x_n < x} c_n \quad (a < x < b). \quad (4.31)$$

The summation is to be understood as follows: Sum over those indices n for which $x_n < x$. If there are no points x_n to the left of x , the sum is empty; following the usual convention, we define it to be zero. Since (4.31) converges absolutely, the order in which the terms are arranged is immaterial.²⁴

We leave the verification of the following properties of f to the reader:

- (a) f is monotonically increasing on (a, b) ;
- (b) f is discontinuous at every point of E ; in fact,

$$f(x_n+) - f(x_n-) = c_n.$$

- (c) f is continuous at every other point of (a, b) .

Moreover, it is not hard to see that $f(x-) = f(x)$ at all points of (a, b) . If a function satisfies this condition, we say that f is *continuous from the left*. If the summation in (4.31) were taken over all indices n for which $x_n \leq x$, we would have $f(x+) = f(x)$ at every point of (a, b) ; that is, f would be *continuous from the right*.

Functions of this sort can also be defined by another method; for an example we refer to Theorem 6.16.

INFINITE LIMITS AND LIMITS AT INFINITY

THE EXTENDED REALS, CASHED IN | Two definitions and a theorem, closing the chapter by making $\lim_{x \rightarrow \infty}$ and “ $f(t) \rightarrow +\infty$ ” legitimate. The device is the one hinted at in 1.23: give $\pm\infty$ *neighbourhoods*, and Definition 4.1 restates itself with neighbourhoods in place of distances — no new theory needed, and 4.34 lists which limit laws survive, with precisely the exceptions catalogued as un-

23 A counting proof with real geometry in it: by 4.29 every discontinuity of an increasing f is a jump, the jump intervals $(f(x-), f(x+))$ are pairwise disjoint by (4.26), and disjoint intervals can each be tagged with their own rational — density of $\mathbb{Q}$ (1.20(b)) supplies the tag, countability of $\mathbb{Q}$ (2.13) does the counting. The same disjoint-intervals-are-countable argument proved Exercise 2.29. (Choosing $r(x)$ for each x is another silent appeal to choice; here it can be avoided — take the rational with least index in a fixed enumeration.)

24 Chapter 3’s rearrangement theorem (3.55) quietly holding up the definition: the indices with $x_n < x$ arrive in no particular order, so the sum is well defined only because the convergence is absolute. A conditionally convergent (c_n) would make (4.31) meaningless. This jump function is the prototype for the integrators of Chapter 6 (Rudin points to 6.16), where its jumps become point masses.

defined at 1.23.

To enable us to operate in the extended real number system, we shall now enlarge the scope of Definition 4.1 by reformulating it in terms of neighborhoods.

For any real number x , we have already defined a neighborhood of x to be any segment $(x - \delta, x + \delta)$.

4.32 Definition. For any real c , the set of real numbers x such that $x > c$ is called a *neighborhood of $+\infty$* and is written $(c, +\infty)$. Similarly, the set $(-\infty, c)$ is a neighborhood of $-\infty$.

4.33 Definition. Let f be a real function defined on $E \subset \mathbb{R}$. We say that

$$f(t) \rightarrow A \text{ as } t \rightarrow x,$$

where A and x are in the extended real number system, if for every neighborhood U of A there is a neighborhood V of x such that $V \cap E$ is not empty, and such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$.²⁵

A moment's consideration will show that this coincides with Definition 4.1 when A and x are real.

The analogue of Theorem 4.4 is still true, and the proof offers nothing new. We state it, for the sake of completeness.

4.34 Theorem. Let f and g be defined on $E \subset \mathbb{R}$. Suppose

$$f(t) \rightarrow A, \quad g(t) \rightarrow B \quad \text{as } t \rightarrow x.$$

Then

- (a) $f(t) \rightarrow A'$ implies $A' = A$,
- (b) $(f + g)(t) \rightarrow A + B$,
- (c) $(fg)(t) \rightarrow AB$,
- (d) $(f/g)(t) \rightarrow A/B$,

provided the right members of (b), (c), and (d) are defined.

Note that $\infty - \infty$, $0 \cdot \infty$, ∞ / ∞ , and $A/0$ are not defined (see Definition 1.23).

25 One definition now covers nine cases (x and A each finite, $+\infty$, or $-\infty$), because “neighbourhood” absorbs the difference: an ε -ball around a real number, a ray beyond c for $\pm\infty$. This is the topologist’s trick — rephrase metrically-stated definitions in terms of neighbourhoods and they generalise by themselves. Theorem 3.15’s sequence version was the same move.

CHAPTER 4 IN ONE PARAGRAPH

Continuity is one property in three costumes: ε - δ at a point (4.5), preservation of sequential limits (4.2 with 4.6), and open inverse images (4.8) — the last being the form that proves theorems. What continuity transports is the content of the chapter: compactness pushes forward (4.14), and with it boundedness, attained extrema, continuous inverses, and uniform continuity (4.15–4.19); connectedness pushes forward (4.22), and with it the intermediate

value theorem (4.23). The counterexamples 4.20, 4.21, 4.24 and 4.27 mark the exact boundaries: nothing survives the loss of compactness except by accident, and the intermediate value property is weaker than continuity. Monotonic functions are the tame class — one-sided limits always exist (4.29), so only jumps occur, and only countably many (4.30), though possibly on a dense set (4.31). Chapter 6 will integrate against exactly such jump functions, and Chapter 5's derivative inherits every tool built here.

EXERCISES

HOW TO USE THESE

Chapter 4's exercises divide into a uniform-continuity core, a counterexample gallery, and a toolbox of functions used throughout the rest of the book.

The chain that matters. $4.9 \rightarrow 4.11 \rightarrow 4.13$ develops what uniform continuity is really for: it preserves Cauchy sequences, and therefore lets a function be extended from a dense subset to the whole space — the mechanism by which e^x , defined on $\mathbb{Q}$ in Exercise 1.6, becomes a function on $\mathbb{R}$ in Chapter 8.

The toolbox. 4.20–4.22 build the distance function ρ_E and use it to separate closed sets (normality) — machinery reused in Chapters 7 and 10. 4.23–4.24 are convex functions, promised in the box at 2.17.

The gallery. 4.1, 4.7, 4.16, 4.18, 4.21(b), 4.25(b) are counterexamples to remember; 4.18 (Thomae's function) and 4.7 (continuity along every line without continuity) are famous.

Alternative proofs. 4.10 reproves uniform continuity on compact sets by sequences — compare it with the cover proof of 4.19 to see the two faces of compactness side by side.

26 No. The condition compares the two sides with each other, never with $f(x)$ — so any function with matching one-sided limits but the wrong value passes: $f = 0$ except $f(0) = 1$. A symmetric condition cannot see an asymmetric failure. First lesson of the set: check what a limit condition actually quantifies over.

27 Pull back instead of pushing forward: $f(E)$ is closed, so $f^{-1}(f(E))$ is closed (corollary to 4.8) and contains E , hence contains $\bar{E}$. For proper inclusion, let the closure point escape to infinity: $f(x) = 1/(1+x^2)$ on $E = \mathbb{N}$ gives $f(\bar{E}) = f(E)$ but $0 \in f(\bar{E})$. This is the exercise promised in Chapter 2's box on why f^{-1} outranks f .

4.1. Suppose f is a real function defined on $\mathbb{R}^1$ which satisfies

$$\lim_{h \rightarrow 0} [f(x+h) - f(x-h)] = 0$$

for every $x \in \mathbb{R}^1$. Does this imply that f is continuous?²⁶

4.2. If f is a continuous mapping of a metric space X into a metric space Y , prove that

$$f(\bar{E}) \subset \overline{f(E)}$$

for every set $E \subset X$. ($\bar{E}$ denotes the closure of E .) Show, by an example, that $f(\bar{E})$ can be a proper subset of $\overline{f(E)}$.²⁷

- 4.3.** Let f be a continuous real function on a metric space X . Let $Z(f)$ (the *zero set* of f) be the set of all $p \in X$ at which $f(p) = 0$. Prove that $Z(f)$ is closed.²⁸
- 4.4.** Let f and g be continuous mappings of a metric space X into a metric space Y , and let E be a dense subset of X . Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for all $p \in E$, prove that $g(p) = f(p)$ for all $p \in X$. (In other words, a continuous function is determined by its values on a dense subset of its domain.)²⁹
- 4.5.** If f is a real continuous function defined on a closed set $E \subset \mathbb{R}^1$, prove that there exist continuous real functions g on $\mathbb{R}^1$ such that $g(x) = f(x)$ for all $x \in E$. (Such functions g are called *continuous extensions* of f from E to $\mathbb{R}^1$.) Show that the result becomes false if the word “closed” is omitted. Extend the result to vector-valued functions. *Hint:* Let the graph of g be a straight line on each of the segments which constitute the complement of E (compare Exercise 2.29). The result remains true if $\mathbb{R}^1$ is replaced by any metric space, but the proof is not so simple.³⁰
- 4.6.** If f is defined on E , the *graph* of f is the set of points $(x, f(x))$, for $x \in E$. In particular, if E is a set of real numbers and f is real-valued, the graph of f is a subset of the plane.
Suppose E is compact, and prove that f is continuous on E if and only if its graph is compact.³¹
- 4.7.** If $E \subset X$ and if f is a function defined on X , the *restriction* of f to E is the function g whose domain of definition is E , such that $g(p) = f(p)$ for $p \in E$. Define f and g on $\mathbb{R}^2$ by: $f(0,0) = g(0,0) = 0$, $f(x,y) = xy^2/(x^2 + y^4)$, $g(x,y) = xy^2/(x^2 + y^6)$ if $(x,y) \neq (0,0)$. Prove that f is bounded on $\mathbb{R}^2$, that g is unbounded in every neighborhood of $(0,0)$, and that f is not continuous at $(0,0)$; nevertheless, the restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous!³²
- 4.8.** Let f be a real uniformly continuous function on the bounded set E in $\mathbb{R}^1$. Prove that f is bounded on E . Show that the conclusion is false if the boundedness of E is omitted from the hypothesis.³³
- 4.9.** Show that the requirement in the definition of uniform continuity can be rephrased as follows, in terms of diameters of sets: To every $\varepsilon > 0$ there exists a $\delta > 0$ such that $\text{diam } f(E) < \varepsilon$ for all $E \subset X$ with $\text{diam } E < \delta$.³⁴
- 4.10.** Complete the details of the following alternative proof of Theo-

28 One line with the right tool:

$Z(f) = f^{-1}(\{0\})$, and $\{0\}$ is closed.

The 4.8 corollary is doing what it was built for — and 4.22 below will prove the converse, that every closed set is some function’s zero set.

29 The rigidity principle. For the second part, take $p \in X$, approach it from E (density plus 3.2(d)), and let continuity carry the agreement to the limit. It is why knowing a continuous function on $\mathbb{Q}$ determines it on $\mathbb{R}$ — the fact 4.13 will exploit for extensions, and Chapter 8 for e^x .

30 Exercise 2.29 is the structural input: the open complement of E is a countable union of disjoint segments, and bridging each with a chord leaves continuity intact. Without closedness the failure is immediate — $1/x$ on $(0,1)$ has nowhere to send 0. The vector-valued case is componentwise (4.10).

31 A characterisation worth carrying: for compact domains, continuity of the function equals compactness of the graph. The forward direction is 4.14 applied to $x \mapsto (x, f(x))$ (continuous by 4.10). The converse is where the work is — from a sequence $x_n \rightarrow x$, compactness of the graph forces a convergent subsequence of $(x_n, f(x_n))$ whose limit must be $(x, f(x))$, and 3.20-style bookkeeping upgrades subsequences to the sequence. Graphs recur at 4.7’s pathologies and in Chapter 9’s implicit function theorem.

32 The classic warning for Chapter 9: continuity along every line through a point does not give continuity at the point. Approach f along the parabola $x = y^2$ and it sits at $\frac{1}{2}$; along lines it dies to 0. Directions are not enough in dimension two — curves can sneak in. Remember this when partial derivatives appear (9.20): their existence is a line-by-line statement, and this is why it proves so little.

33 Uniform continuity buys finitely many δ -steps across a bounded set: cover $\overline{E}$ (compact!) by finitely many δ -balls, or argue directly — each step changes f by at most ε , and boundedness of E caps the number of steps. Compare 4.20(a): plain continuity could not do this. For the failure, $f(x) = x$ on $\mathbb{R}$.

34 Uniform continuity as a statement about sets, with the points quantified away — 3.9's diameter doing for functions what it did for Cauchy sequences. This is the form the extension theorem 4.13 runs on: small sets have small images, uniformly.

35 Worth writing out in full, as the mirror image of 4.19's cover proof. Negate the uniform statement (one bad ε , witnesses for every $\delta_n = 1/n$ — the same contrapositive harvest as 4.2), compactify: some subsequence $p_{n_k} \rightarrow p$, drag q_{n_k} along since the distance dies, and continuity at the single point p contradicts the standing gap ε . Covers versus sequences is the same fork as 2.32 versus 2.41(c) — Rudin chose covers for the text, Tao's compactness definition would make this the primary proof.

36 The heart of the uniform/pointwise distinction: a Cauchy sequence's terms crowd each other at unpredictable locations, so only a location-independent δ can convert crowding in X to crowding in Y . Plain continuity fails the task: $f(x) = 1/x$ on $(0, 1)$ maps the Cauchy sequence $(1/n)$ to the divergent (n) .

37 The relay of 4.7 again, with both contracts now point-free: ε buys η from g uniformly, η buys δ from f uniformly. Nothing in the composition argument ever used the base point — which is the observation this exercise wants made.

rem 4.19: If f is not uniformly continuous, then for some $\varepsilon > 0$ there are sequences $(p_n), (q_n)$ in X such that $d_X(p_n, q_n) \rightarrow 0$ but $d_Y(f(p_n), f(q_n)) > \varepsilon$. Use Theorem 2.37 to obtain a contradiction.³⁵

4.11. Suppose f is a uniformly continuous mapping of a metric space X into a metric space Y , and prove that $(f(x_n))$ is a Cauchy sequence in Y for every Cauchy sequence (x_n) in X . Use this result to give an alternative proof of the theorem stated in Exercise 4.13.³⁶

4.12. A uniformly continuous function of a uniformly continuous function is uniformly continuous. State this more precisely and prove it.³⁷

4.13. Let E be a dense subset of a metric space X , and let f be a uniformly continuous real function defined on E . Prove that f has a continuous extension from E to X (see Exercise 4.5 for terminology). (Uniqueness follows from Exercise 4.4.) *Hint:* For each $p \in X$ and each positive integer n , let $V_n(p)$ be the set of all $q \in E$ with $d(p, q) < 1/n$. Use Exercise 4.9 to show that the intersection of the closures of the sets $f(V_1(p)), f(V_2(p)), \dots$ consists of a single point, say $g(p)$, of $\mathbb{R}^1$. Prove that the function g so defined on X is the desired extension of f .

Could the range space $\mathbb{R}^1$ be replaced by $\mathbb{R}^k$? By any compact metric space? By any complete metric space? By any metric space?³⁸

4.14. Let $I = [0, 1]$ be the closed unit interval. Suppose f is a continuous mapping of I into I . Prove that $f(x) = x$ for at least one $x \in I$.³⁹

4.15. Call a mapping of X into Y *open* if $f(V)$ is an open set in Y whenever V is an open set in X . Prove that every continuous open mapping of $\mathbb{R}^1$ into $\mathbb{R}^1$ is monotonic.⁴⁰

4.16. Let $[x]$ denote the largest integer contained in x , that is, $[x]$ is the integer such that $x - 1 < [x] \leq x$, and let $(x) = x - [x]$ denote the fractional part of x . What discontinuities do the functions $[x]$ and (x) have?⁴¹

4.17. Let f be a real function defined on (a, b) . Prove that the set of points at which f has a simple discontinuity is at most countable. *Hint:* Let E be the set on which $f(x-) < f(x+)$. With each point x of E , associate a triple (p, q, r) of rational numbers such that

- (a) $f(x-) < p < f(x+)$,
- (b) $a < q < t < x$ implies $f(t) < p$,
- (c) $x < t < r < b$ implies $f(t) > p$.

The set of all such triples is countable. Show that each triple is associated with at most one point of E . Deal similarly with the other possible types of simple discontinuities.⁴²

- 4.18.** Every rational x can be written in the form $x = m/n$, where $n > 0$, and m and n are integers without any common divisors. When $x = 0$, we take $n = 1$. Consider the function f defined on $\mathbb{R}^1$ by

$$f(x) = \begin{cases} 0 & (x \text{ irrational}), \\ \frac{1}{n} & \left(x = \frac{m}{n}\right). \end{cases}$$

Prove that f is continuous at every irrational point, and that f has a simple discontinuity at every rational point.⁴³

- 4.19.** Suppose f is a real function with domain $\mathbb{R}^1$ which has the intermediate value property: If $f(a) < c < f(b)$, then $f(x) = c$ for some x between a and b . Suppose also, for every rational r , that the set of all x with $f(x) = r$ is closed. Prove that f is continuous.

Hint: If $x_n \rightarrow x_0$ but $f(x_n) > r > f(x_0)$ for some r and all n , then $f(t_n) = r$ for some t_n between x_0 and x_n ; thus $t_n \rightarrow x_0$. Find a contradiction. (N. J. Fine, *Amer. Math. Monthly*, vol. 73, 1966, p. 782.)⁴⁴

- 4.20.** If E is a nonempty subset of a metric space X , define the distance from $x \in X$ to E by

$$\rho_E(x) = \inf_{z \in E} d(x, z).$$

- (a) Prove that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.
- (b) Prove that ρ_E is a uniformly continuous function on X , by showing that

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y)$$

for all $x \in X, y \in X$.

Hint: $\rho_E(x) \leq d(x, z) \leq d(x, y) + d(y, z)$, so that

$$\rho_E(x) \leq d(x, y) + \rho_E(y).$$

[38] The most consequential exercise of the chapter. The hint is 3.10(b) recycled: shrinking sets with shrinking diameters (by 4.9!) whose closures nest — their intersection is the value the extension must take. The closing questions locate the true hypothesis: everything works in any complete range space (the nested-closed-sets argument is Exercise 3.21), and completeness is exactly what is needed — the limit must have somewhere to live. This machine builds e^x from its rational values (Chapter 8) and, in functional analysis, extends densely-defined operators.

39 The one-dimensional fixed point theorem, and the standard first application of 4.23: $g(x) = f(x) - x$ has $g(0) \geq 0$, $g(1) \leq 0$, so the intermediate value theorem plants a zero. One line — but only because both endpoints are trapped in I . Brouwer's theorem is this statement in $\mathbb{R}^k$, and needs far heavier tools.

40 Contrapositive: a continuous non-monotonic function has an interior extremum on some $[a, b]$ (attained by 4.16), and then $f((a, b))$ contains its own maximum — a closed endpoint — so is not open. Openness of images turns out to be a strong, rigidity-inducing property; compare the box at 2.2 on why it is not the definition of continuity.

41 Simple jumps at every integer, of size 1, both functions continuous from the right — the floor function is 4.31's jump construction in its simplest form, with $E = \mathbb{Z}$ and all $c_n = 1$. It returns as the integrator of choice in Chapter 6 (6.16), where its jumps turn integrals into sums.

42 4.30's tagging trick without monotonicity: one rational is no longer enough (jumps can overlap), so each jump is tagged with a triple — its level plus a rational buffer on each side — and 2.13 counts triples. The general principle: to prove a set countable, inject it into $\mathbb{Q}^n$; the injectivity is where the analysis lives.

43 Thomae's function — continuous exactly on the irrationals. The key: near any point, rationals with small denominators are scarce, so $f(t) < \varepsilon$ once t avoids the finitely many fractions with $n \leq 1/\varepsilon$ in a bounded window. The asymmetry with 4.27(b) is instructive, and one can ask for the reverse: a function continuous exactly on $\mathbb{Q}$. None exists — the continuity set of any function is a countable intersection of open sets, and $\mathbb{Q}$ is not one, by Baire (Exercise 3.22). That is a satisfying loop: Baire's theorem answering a question about a single function.

44 The repaired converse of 4.23: the intermediate value property alone fails (4.24, 4.27(d)), but adding closed rational level sets restores continuity. In the hint, the level t_n 's accumulate at x_0 , so closedness forces $f(x_0) = r$ — contradiction. Note how the hypothesis is used only through sequences: the corollary to 4.8 in its sequential clothing.

45 The workhorse function of metric topology. Part (a) says ρ_E sees exactly the closure; part (b) says it is 1-Lipschitz — the strongest, cleanest form of uniform continuity, with $\delta = \varepsilon$. The hint's move (bound by an over-estimate, then take the inf) is how one always handles infima of distances.

46 4.16 and 4.20 cooperating: ρ_F is positive on K (by 4.20(a), since F is closed and disjoint), and a positive continuous function on a compact set attains a positive minimum. For the failure take $F = \mathbb{Z}$ and $K = \{n + 1/n\}$: disjoint, closed, distance 0. Two closed sets can approach each other at infinity; a compact set has no infinity to escape to.

47 Urysohn's lemma for metric spaces, built from Exercise 4.20's bricks in one display. The denominator never vanishes — ρ_A and ρ_B cannot both be zero, by 4.20(a) and disjointness — and the two preimages under f separate A from B openly, with 4.8 certifying openness. In general topology normality is hard-won; metrics give it for free, which is one reason this book never needs spaces beyond metric ones.

4.21. Suppose K and F are disjoint sets in a metric space X , K is compact, F is closed. Prove that there exists $\delta > 0$ such that $d(p, q) > \delta$ if $p \in K$, $q \in F$. Hint: ρ_F is a continuous positive function on K .

Show that the conclusion may fail for two disjoint closed sets if neither is compact.⁴⁶

4.22. Let A and B be disjoint nonempty closed sets in a metric space X , and define

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)} \quad (p \in X).$$

Show that f is a continuous function on X whose range lies in $[0, 1]$, that $f(p) = 0$ precisely on A and $f(p) = 1$ precisely on B . This establishes a converse of Exercise 4.3: Every closed set $A \subset X$ is $Z(f)$ for some continuous real f on X . Setting

$$V = f^{-1}\left(\left[0, \frac{1}{2}\right)\right), \quad W = f^{-1}\left(\left(\frac{1}{2}, 1\right]\right),$$

show that V and W are open and disjoint, and that $A \subset V$, $B \subset W$. (Thus pairs of disjoint closed sets in metric spaces can be covered by pairs of disjoint open sets. This property of metric spaces is called *normality*.)⁴⁷

4.23. A real-valued function f defined in (a, b) is said to be *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

whenever $a < x < b$, $a < y < b$, $0 < \lambda < 1$. Prove that every convex function is continuous. Prove that every increasing convex function of a convex function is convex. (For example, if f is convex, so is e^f .)

If f is convex in (a, b) and if $a < s < t < u < b$, show that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t}.$$

48

4.24. Assume that f is a continuous real function defined in (a, b) such that

$$f\left(\frac{x + y}{2}\right) \leq \frac{f(x) + f(y)}{2}$$

for all $x, y \in (a, b)$. Prove that f is convex.⁴⁹

4.25. If $A \subset \mathbb{R}^k$ and $B \subset \mathbb{R}^k$, define $A + B$ to be the set of all sums $\mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in A$, $\mathbf{y} \in B$.

- (a) If K is compact and C is closed in $\mathbb{R}^k$, prove that $K + C$ is closed.

Hint: Take $\mathbf{z} \notin K + C$, put $F = \mathbf{z} - C$, the set of all $\mathbf{z} - \mathbf{y}$ with $\mathbf{y} \in C$. Then K and F are disjoint. Choose δ as in Exercise 4.21. Show that the open ball with center $\mathbf{z}$ and radius δ does not intersect $K + C$.

- (b) Let α be an irrational real number. Let C_1 be the set of all integers, let C_2 be the set of all $n\alpha$ with $n \in C_1$. Show that C_1 and C_2 are closed subsets of $\mathbb{R}^1$ whose sum $C_1 + C_2$ is not closed, by showing that $C_1 + C_2$ is a countable dense subset of $\mathbb{R}^1$.⁵⁰

4.26. Suppose X, Y, Z are metric spaces, and Y is compact. Let f map X into Y , let g be a continuous one-to-one mapping of Y into Z , and put $h(x) = g(f(x))$ for $x \in X$. Prove that f is uniformly continuous if h is uniformly continuous. *Hint:* g^{-1} has compact domain $g(Y)$, and $f(x) = g^{-1}(h(x))$.

Prove also that f is continuous if h is continuous. Show (by modifying Example 4.21, or by finding a different example) that the compactness of Y cannot be omitted from the hypotheses, even when X and Z are compact.⁵¹

48 The slope inequality at the end is the real content — chords steepen from left to right — and continuity falls out of it: near any interior point the difference quotients are wedged between two fixed chords' slopes, so f is locally Lipschitz. Note "interior" is essential; on $[a, b]$ a convex function may jump at the endpoints. These inequalities return for the differentiable case in Chapter 5's exercises.

49 Midpoint convexity plus continuity gives full convexity: iterate midpoints to reach all dyadic $\lambda = k/2^n$, then let continuity fill in the rest — density doing its usual work (Exercise 4.4's principle applied to λ). Without continuity the statement fails, but the counterexamples need a Hamel basis, far beyond this book.

50 Part (b) is the famous fact that $\{m + n\alpha\}$ is dense for irrational α — Weyl's equidistribution in embryo. The proof is pigeonhole: the fractional parts $(n\alpha)$ are infinitely many points in $[0, 1)$, two must fall within ε , and their difference generates an ε -mesh. Against (a), it shows compactness of one summand is not a removable convenience: closed + closed can be dense-and-not-closed.

51 A cancellation theorem: composing with an injective continuous g can be undone — because 4.17 makes g^{-1} continuous, indeed uniformly so by 4.19, and 4.12 composes the uniformities. Every hypothesis of this chapter's compactness section is consumed in one exercise; the counterexample without compact Y is 4.21's wrap-around map playing the role of g .

Appendix to Chapter 4

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 4 is a chapter about *structure*: what continuity does to compactness and connectedness. He buys that generality by postponing every concrete function — $\sin$, $\exp$, $\log$, x^α do not exist until Chapter 8 — and by leaving the working criteria for limits and uniform continuity to the exercises. The Chinese and Tao texts make the opposite trade. Reading them alongside supplies five things Rudin's chapter genuinely lacks:

- 1. Criteria (A).** A Cauchy criterion for function limits; one definition of limit that specialises to all of Rudin's; a squeeze law; and oscillation, which measures *how* discontinuous a function is at a point.
- 2. Asymptotics (B).** The o/O calculus and equivalent infinitesimals — the machinery every later computation runs on, and which Rudin never introduces at all.
- 3. The postponed limits (C).** $\lim_{x \rightarrow 0} (\sin x)/x = 1$ and $\lim_{x \rightarrow \infty} (1 + 1/x)^x = e$, with the proofs the other books give.
- 4. Tools left in Rudin's exercises (D, E).** Inverse functions without compactness; the sequential criterion for uniform continuity; gluing; extension to the closure. Rudin asks for several of these as Exercises 4.11–4.13; the other texts prove them in the running text and then use them.
- 5. Finer structure (F).** Rudin's two kinds of discontinuity refine to four; connectedness gains a path-connected cousin and a decomposition into components.

How to read it. None of this is needed to follow Rudin, which is why it sits in an appendix rather than in his chapter. Sections C and D in particular are *logically* out of order in his development — they use functions he has not built yet — and each says so where it matters. Read the appendix as a second

pass over Chapter 4, or as the answer to “how would I actually compute with this?” The exercises begin on the last section.

NOT IN RUDIN

SUPPLEMENT A — CRITERIA FOR LIMITS THAT RUDIN OMMITS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Definition 4.1 defines $\lim_{x \rightarrow p} f(x) = q$ by ε - δ , and 4.2 characterises it by sequences: $f(p_n) \rightarrow q$ for every sequence $p_n \rightarrow p$ in E with $p_n \neq p$.

The gap. Both forms presuppose the candidate q . Neither can be used to decide whether a limit *exists* when no value suggests itself, and Rudin supplies no criterion that does — no Cauchy condition, no squeeze, and no way of measuring how badly a limit fails to exist.

Where the gap bites. At Exercise 4.13, where a uniformly continuous function must be extended to the closure and the limit at each new point has to be produced from nothing; Rudin’s hint amounts to proving a Cauchy criterion by hand in that special case. It bites again at 4.26–4.28, where discontinuities are classified but never quantified, so “how discontinuous” has no answer.

What this section supplies. The Cauchy criterion for function limits; one definition of limit that specialises to all four of Rudin’s; the squeeze law he uses but never states; and oscillation, which turns “continuous at p ” into the numerical statement $\omega_f(p) = 0$ and makes the set of continuity points a computable object.

S4.1 Theorem. (*Cauchy criterion for function limits.*) Let X be a metric space, $E \subset X$, p a limit point of E , and $f : E \rightarrow Y$ with Y complete (for instance $Y = \mathbb{R}^k$). Then $\lim_{x \rightarrow p} f(x)$ exists if and only if: for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$d_Y(f(x'), f(x'')) < \varepsilon$$

for all $x', x'' \in E$ with $0 < d_X(x', p) < \delta$ and $0 < d_X(x'', p) < \delta$.

SOURCE: ECNU 定理 3.11 (bk. p. 52); USTC 定理 9 (bk. pp. 29–30). Rudin has no counterpart in Chapter 4.

Proof sketch. *Necessity* is the triangle inequality: if $f(x) \rightarrow q$, take δ for $\varepsilon/2$ and add.

Sufficiency routes through sequences, exactly as 4.2 does. Let (p_n)

be any sequence in $E \setminus \{p\}$ with $p_n \rightarrow p$. The condition makes $(f(p_n))$ a Cauchy sequence, hence convergent by completeness (3.11). If (p'_n) is a second such sequence with a different limit, interleave the two into one sequence; its image is Cauchy, so all its subsequential limits agree, and the two limits coincide. Now 4.2 applies. $\square$

Why Rudin can do without it, and you cannot

Rudin never needs S4.1 because he almost never has to *produce* a function limit: his theorems take a convergent hypothesis and transport it. The one place he does need it, he writes it out by hand — Exercise 4.13, extending a uniformly continuous function to the closure, is precisely “Cauchy criterion + completeness” done in a special case. Supplement E below (S4.15) states the general fact the exercise is a shadow of. Note what the criterion costs: completeness of the *target*. Over $\mathbb{Q}$ it is false, and the failure is the same one that motivates all of Chapter 1.

S4.2 Definition. (*Limits along a subset.*) Let $E \subset X$, let p be adherent to E — that is, $p \in \overline{E}$ — and let f be defined on E . Write

$$\lim_{x \rightarrow p, x \in E} f(x) = q$$

if for every $\varepsilon > 0$ there is $\delta > 0$ with $d(f(x), q) < \varepsilon$ for all $x \in E$ satisfying $d(x, p) < \delta$.

SOURCE: Tao I, Def. 9.3.6 (bk. p. 221), with adherent points Def. 9.1.8.

ONE DEFINITION IN PLACE OF RUDIN'S FOUR

Tao's single notion specialises to everything Rudin defines separately. Fix a function f on a set D and a point p :

Take $E =$	and you get
$D \setminus \{p\}$	Rudin's $\lim_{x \rightarrow p} f(x)$ (4.1)
$D \cap (p, \infty)$	the right-hand limit $f(p+)$ (4.25)
$D \cap (-\infty, p)$	the left-hand limit $f(p-)$
D itself	continuity at p , when the value is $f(p)$
D , with $p = +\infty$	$\lim_{x \rightarrow +\infty} f(x)$ (4.33)
$D \cap A$	the limit of the restriction $f _A$ (4.12)

The fourth row is the interesting one. Rudin's limit at p is *deleted*: p is thrown out of the domain, which is why 4.1 requires p to be a limit point and why $\lim_{x \rightarrow p} f(x)$ can differ from $f(p)$ — the whole content of Definition 4.5 and of the discontinuity classification. Tao's undeleted version with $E = D$ makes “continuous at p ” literally the statement “the limit along the domain is $f(p)$.” The two conventions disagree, and each is chosen to make its own theorems cleanest; knowing which one is in force is a standing source of confusion between textbooks.

S4.3 Theorem. (*Squeeze.*) If $f \leq g \leq h$ on E and $\lim_{x \rightarrow p, x \in E} f = \lim_{x \rightarrow p, x \in E} h = L$, then $\lim_{x \rightarrow p, x \in E} g = L$.

SOURCE: Tao I, Ex. 9.3.5; ECNU 定理 3.2 (bk. p. 46).

Proof. Let (p_n) be any sequence in E with $p_n \rightarrow p$. Then $f(p_n) \rightarrow L$ and $h(p_n) \rightarrow L$, and $f(p_n) \leq g(p_n) \leq h(p_n)$, so $g(p_n) \rightarrow L$ by the sequence squeeze (3.19 applied to $g(p_n) - f(p_n) \leq h(p_n) - f(p_n)$). Since the sequence was arbitrary, 4.2 gives the conclusion. $\square$

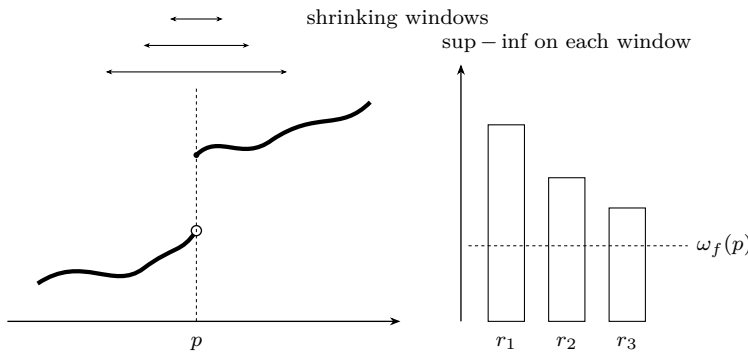
Rudin proves the sequence version (3.19 is its cousin) and never states the function version, though he uses the reasoning freely.⁵²

S4.4 Definition. (*Oscillation.*) For f real and bounded near p , put

$$\omega_f(p) = \lim_{r \rightarrow 0^+} \left[\sup_{d(x,p) < r} f(x) - \inf_{d(x,p) < r} f(x) \right].$$

The bracket decreases as r decreases, so the limit exists.

SOURCE: Tao I, Ex. 9.3.4 and Tao II, Ex. 2.2.9 (as $\limsup - \liminf$); Yu Pin, Lecture 21 Lemma 131 (p. 224), who defines it for maps of metric spaces as $\omega(f, p) = \inf_{U \ni p} \text{diam } f(U)$ over open U . USTC and ECNU introduce 振幅 only later, for integrability.



Oscillation: measure $\sup - \inf$ over a window about p and let the window close. The bars fall but stall at the size of the jump. The function is continuous at p exactly when they fall to 0.

S4.5 Theorem. f is continuous at p if and only if $\omega_f(p) = 0$. Moreover $\Omega_\varepsilon = \{x : \omega_f(x) \geq \varepsilon\}$ is closed for each $\varepsilon > 0$; consequently the set of points at which f is discontinuous is $\bigcup_n \Omega_{1/n}$, a countable union of closed sets, and the set where f is continuous is a countable intersection of open sets.

SOURCE: Yu Pin, Lemmas 131–132 and p. 227, for arbitrary maps of metric spaces; Tao I, Ex. 9.3.4.

Proof. If $\omega_f(p) = 0$ then given $\varepsilon > 0$ some window has $\sup - \inf < \varepsilon$, so $|f(x) - f(p)| < \varepsilon$ throughout it: f is continuous at p . Conversely continuity makes both $\sup$ and $\inf$ over a small enough window lie within ε of $f(p)$, so their difference is at most 2ε .

For the second claim, let $\omega_f(x_0) < \varepsilon$. Then some ball $B(x_0, r)$ has $\sup - \inf < \varepsilon$ on it, and every $x \in B(x_0, r)$ has a smaller ball inside

⁵² Cheap to prove and constantly needed — Supplement C below cannot get off the ground without it. The mechanism is 4.2: pull back to sequences and use the sequence squeeze.

$B(x_0, r)$, so $\omega_f(x) < \varepsilon$ too. Hence $\{\omega_f < \varepsilon\}$ is open and its complement Ω_ε is closed. Finally f is discontinuous at x exactly when $\omega_f(x) > 0$, that is when $x \in \Omega_{1/n}$ for some n . $\square$

The payoff: why Thomae's function has no mirror image

Rudin's Exercise 4.18 produces a function continuous at every irrational and discontinuous at every rational. The obvious next question — can a function be continuous at every *rational* and discontinuous at every irrational? — Rudin never asks, and with his Chapter 4 tools he cannot answer it. S4.5 answers it: no.

The continuity set is always $\bigcap_n \{x : \omega_f(x) < 1/n\}$, an intersection of countably many open sets. If some f were continuous exactly on $\mathbb{Q}$, then $\mathbb{Q}$ would be such an intersection, say $\mathbb{Q} = \bigcap_n U_n$ with each U_n open and dense (each contains $\mathbb{Q}$). Enumerate $\mathbb{Q} = \{q_1, q_2, \dots\}$; then

$$\emptyset = \left(\bigcap_n U_n \right) \cap \left(\bigcap_k (\mathbb{R} \setminus \{q_k\}) \right)$$

exhibits $\emptyset$ as a countable intersection of dense open sets, which Baire's theorem forbids. So Thomae's function has no mirror image, and the asymmetry between $\mathbb{Q}$ and its complement — the theme running from 2.44 to Chapter 3's rearrangements — shows up once more.

Rudin proves Baire's theorem in Exercise 3.22; this is one of its first real uses.

NOT IN RUDIN

SUPPLEMENT B — ORDERS OF MAGNITUDE: THE CALCULUS OF LIMITS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Nothing — and that is the point. There is no o , no O , no notion of one quantity being negligible beside another anywhere in Chapters 1–4. Limits are proved to exist and are then named.

The gap. Analysis outside this book is conducted almost entirely in asymptotic notation, and the notation is not decoration: it is how one keeps track of which error terms matter. Rudin's abstract line never needs it because he never computes a limit that is not already a theorem.

Where the gap bites. At 5.15, where Taylor's theorem is really a statement that a remainder is $o((x-a)^n)$ and has to be phrased instead with an explicit intermediate point; at 3.20, where the special limits are exactly statements about relative

rates of growth; and in every computation a reader will meet outside these covers.

What this section supplies. The definitions, the substitution theorem that makes them usable, and — most importantly — the pitfall that substitution is legal in factors and illegal in summands, which is the error the notation invites.

S4.6 Definition. Let f, g be defined near p (with $g \neq 0$ there). As $x \rightarrow p$:

- (a) $f = o(g)$ means $f/g \rightarrow 0$;
- (b) $f = O(g)$ means $|f| \leq C|g|$ near p , for some constant C ;
- (c) $f \sim g$ (equivalent) means $f/g \rightarrow 1$.

SOURCE: ECNU 无穷小量的阶, bk. pp. 57–59; USTC 等价无穷小替换, bk. pp. 32–35.

The equality sign in $f = o(g)$ is an abuse: $o(g)$ names a *class* of functions and the statement is membership.⁵³

S4.7 Theorem. (Substitution of equivalents.) Suppose $\alpha \sim \beta$ as $x \rightarrow p$. If $\lim \alpha u$ exists, then $\lim \beta u$ exists and they are equal; if $\lim v/\alpha$ exists, then $\lim v/\beta$ exists and they are equal.

SOURCE: ECNU 定理 3.12; USTC 定理 12.

Proof. Write $\beta = \alpha \cdot (\beta/\alpha)$ where $\beta/\alpha \rightarrow 1$. Then $\beta u = (\alpha u)(\beta/\alpha)$, and the limit of a product is the product of the limits (4.4), so $\lim \beta u = (\lim \alpha u) \cdot 1$. The quotient case is identical with $v/\beta = (v/\alpha)(\alpha/\beta)$. $\square$

SUBSTITUTION IS LEGAL IN PRODUCTS AND QUOTIENTS — NOWHERE ELSE

Since $\tan x \sim x$ and $\sin x \sim x$ as $x \rightarrow 0$, one is tempted to replace both in a *difference*:

$$\tan x - \sin x \stackrel{?}{\sim} x - x = 0,$$

which is not merely imprecise but wrong. The truth is

$$\tan x - \sin x = \sin x \frac{1 - \cos x}{\cos x} \sim x \cdot \frac{x^2}{2} = \frac{x^3}{2},$$

so $(\tan x - \sin x)/\sin^3 x \rightarrow \frac{1}{2}$, not 0. The two equivalences are each accurate to first order, and the difference cancels precisely the order they are accurate to; what survives is third order, about which they say nothing.

The rule to carry: equivalents may be substituted into a *factor* of a product or quotient, never into a summand. When a difference is what you want, expand to enough orders — which is what Taylor's theorem (5.15) will industrialise.

53 Read every “=” involving o or O left-to-right and never reverse it. From $f = o(x)$ and $h = o(x)$ it does not follow that $f = h$; both are merely in the same class. ECNU makes this explicit by writing $o(g) = \{f : f/g \rightarrow 0\}$.

S4.8 Definition. f is an infinitely large quantity as $x \rightarrow p$ if $|f(x)| \rightarrow$

54 Not the same as unbounded, and the difference is the point of the definition. On $(0, \infty)$ the function $x \sin x$ is unbounded, yet it does not tend to ∞ : at $x = n\pi$ it is 0. ECNU isolates this as the standard beginner's error — Rudin's 4.33 defines the notion correctly but never contrasts it with unboundedness.

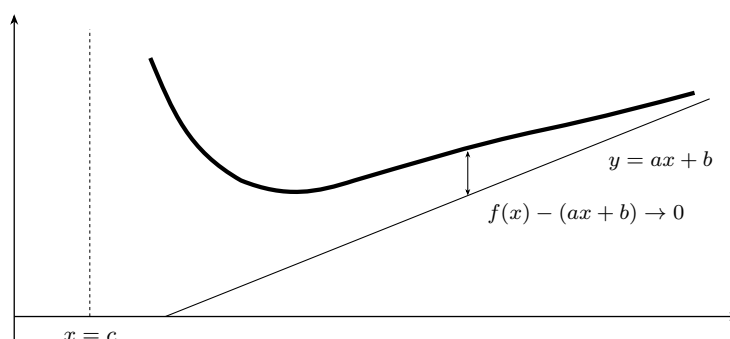
$+\infty$.⁵⁴

S4.9 Definition. (Asymptotes.) A line L is an asymptote of the curve $y = f(x)$ if the distance from $(x, f(x))$ to L tends to 0 as the point recedes to infinity. The line $y = ax + b$ is an asymptote as $x \rightarrow +\infty$ if and only if

$$a = \lim_{x \rightarrow +\infty} \frac{f(x)}{x}, \quad b = \lim_{x \rightarrow +\infty} [f(x) - ax],$$

both limits existing finitely; and $x = c$ is a vertical asymptote if some one-sided limit of f at c is infinite.

SOURCE: ECNU 曲线的渐近线, bk. pp. 61–62; USTC Ex. 1.2.14. Absent from Rudin entirely.



An oblique asymptote. Because the distance from $(x, f(x))$ to the line is $|f(x) - ax - b|/\sqrt{1 + a^2}$, the geometric condition “distance $\rightarrow 0$ ” is the algebraic condition $f(x) - ax - b \rightarrow 0$ — which is why dividing by x recovers a first, and subtracting ax then recovers b . A vertical asymptote is the separate case where f itself blows up.

WORKED EXAMPLE

For $f(x) = \frac{x^3}{x^2 + 2x - 3}$ division gives

$$f(x) = x - 2 + \frac{7x - 6}{x^2 + 2x - 3},$$

and the remainder tends to 0, so $y = x - 2$ is an asymptote in both directions. Since $x^2 + 2x - 3 = (x + 3)(x - 1)$, the lines $x = -3$ and $x = 1$ are vertical asymptotes. Note that the recipe found a and b without any division: $f(x)/x \rightarrow 1$ and $f(x) - x \rightarrow -2$.

NOT IN RUDIN

SUPPLEMENT C — THE TWO LIMITS RUDIN POSTPONES

READ THIS SECTION AS A LOAN AGAINST CHAPTER 8

Rudin has no $\sin$, $\cos$, $\exp$ or $\log$ at this point: they are *defined* in 8.7 by power series, and every property of them is derived there. So the two theorems below are not available in his development, and the

geometric proof of the first uses the area of a circular sector — that is, integration (Chapter 6) and arc length (8.7 again).

They are included because every other text proves them here, because every computation in a first course needs them, and because the arguments are worth seeing. But the logical order is Rudin's, and Yu Pin — who also defines $\sin$ by its series — shows what that costs: he proves the same limit in one line from the estimate

$$\left| \frac{\sin x}{x} - 1 \right| \leq |x|^2 e^{|x|},$$

read straight off the series with no geometry at all. The picture below is then an illustration of a fact already proved, not the proof.

S4.10 Theorem. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$

SOURCE: ECNU 定理 3.14 (bk. p. 53); USTC 定理 10 (bk. p. 30). Rudin obtains it from 8.7.

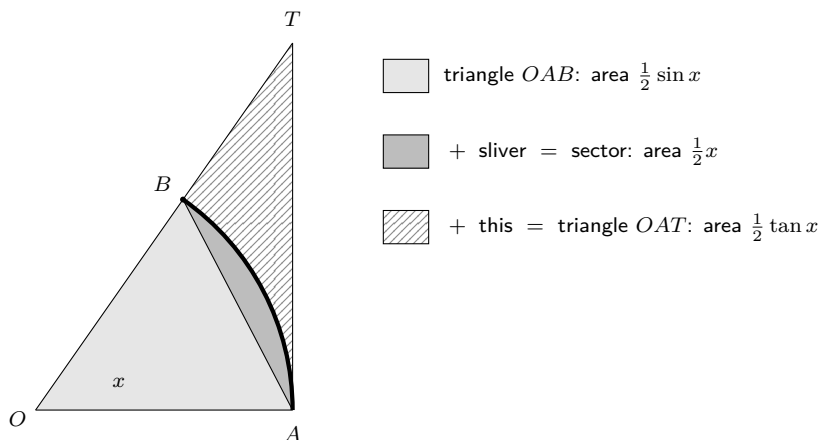
The geometric proof. Let $0 < x < \pi/2$ and work in the unit circle. Comparing the areas of the triangle OAB , the sector OAB , and the triangle OAT gives

$$\frac{1}{2} \sin x < \frac{1}{2} x < \frac{1}{2} \tan x.$$

Dividing by $\frac{1}{2} \sin x > 0$ yields $1 < x / \sin x < 1 / \cos x$, that is

$$\cos x < \frac{\sin x}{x} < 1.$$

Since $\cos x \rightarrow 1$ as $x \rightarrow 0^+$, the squeeze S4.3 gives the limit from the right; the quotient is even, so the two-sided limit follows. $\square$



Three nested regions in the unit circle. Containment of the regions is the chain $\sin x < x < \tan x$ — the only inequality the proof needs. The sector's area being $\frac{1}{2}x$ is what makes radian measure the right measure, and is also what this argument borrows from Chapter 6.

S4.11 Theorem. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$, in both directions; equivalently $\lim_{t \rightarrow 0} (1+t)^{1/t} = e$.

SOURCE: ECNU bk. p. 54; USTC 定理 11. Rudin proves only the sequence case, 3.31.

Proof. Let $x > 0$ and $n = \lfloor x \rfloor$, so $n \leq x < n+1$. Increasing the base increases the power, and increasing the exponent does too, so

$$\left(1 + \frac{1}{n+1}\right)^n < \left(1 + \frac{1}{x}\right)^x < \left(1 + \frac{1}{n}\right)^{n+1}.$$

The left side is $\left(1 + \frac{1}{n+1}\right)^{n+1}$ divided by $\left(1 + \frac{1}{n+1}\right)$ and the right side is $\left(1 + \frac{1}{n}\right)^n$ times $\left(1 + \frac{1}{n}\right)$; by 3.31 both tend to e . Since $n \rightarrow \infty$ as $x \rightarrow +\infty$, the squeeze S4.3 gives the limit. For $x \rightarrow -\infty$ put $x = -(y+1)$ and compute $\left(1 + \frac{1}{x}\right)^x = \left(1 + \frac{1}{y}\right)^{y+1} \rightarrow e$. $\square$

FROM RUDIN'S SEQUENCE TO A CONTINUOUS VARIABLE

Rudin's 3.31 establishes $\left(1 + \frac{1}{n}\right)^n \rightarrow e$ for integers n . To pass to a real variable, trap x between consecutive integers: with $n = \lfloor x \rfloor$ and $x > 0$,

$$\left(1 + \frac{1}{n+1}\right)^n < \left(1 + \frac{1}{x}\right)^x < \left(1 + \frac{1}{n}\right)^{n+1},$$

because increasing the base increases the power and so does increasing the exponent. Both outer terms are the sequence limit multiplied by a factor tending to 1 — the left is $\left(1 + \frac{1}{n+1}\right)^{n+1}$ divided by $\left(1 + \frac{1}{n+1}\right)$, the right is $\left(1 + \frac{1}{n}\right)^n$ times $\left(1 + \frac{1}{n}\right)$ — so both tend to e , and S4.3 closes the gap. The case $x \rightarrow -\infty$ follows from the substitution $x = -(y+1)$.

This bracketing is the standard way to lift *any* sequence limit to a continuous one when the function is monotone between integers, and it is worth keeping for that reason alone.

NOT IN RUDIN

SUPPLEMENT D — MONOTONICITY, INVERSES, AND THE ELEMENTARY FUNCTIONS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Theorem 4.17 proves that a continuous bijection of a *compact* metric space has a continuous inverse, and Example 4.21 shows compactness cannot simply be dropped. Monotone functions appear separately at 4.28–4.30, and the elementary functions do not appear at all — exp, log, sin are built in Chapter 8 from power series.

The gap. On the line there is a second route to the same conclusion that uses no compactness whatever, only order — and

it reaches domains 4.17 cannot, such as open and unbounded intervals. Rudin's two topics, inverse functions and monotone functions, are in fact one topic, and he never connects them.

Where the gap bites. At 4.21 itself, where the circle-wrap counterexample is offered as evidence that compactness is *needed*, when what it really shows is that the domain was not ordered; and at Exercise 4.14, which is about monotone open maps and sits next to a theorem that would explain it.

What this section supplies. The equivalence of injectivity and strict monotonicity for continuous functions on an interval; the inverse function theorem that follows, with no compactness; and the catalogue of elementary functions that this route makes available four chapters early.

S4.12 Theorem. *If f is continuous and one-to-one on an interval $I \subset \mathbb{R}$, then f is strictly monotone on I .*

SOURCE: Tao I, Ex. 9.8.3; USTC 定理 3 states the equivalence (continuous f on $[a, b]$ is invertible iff strictly monotone).

Proof. Suppose f is not strictly monotone. Then there are $u < v < w$ in I with $f(v)$ not strictly between $f(u)$ and $f(w)$ — say $f(v) \geq f(u)$ and $f(v) \geq f(w)$, the other case being symmetric. Pick c with $\max\{f(u), f(w)\} < c < f(v)$, possible unless $f(v)$ equals one of them, which injectivity forbids. The intermediate value theorem 4.23 applied on $[u, v]$ and on $[v, w]$ produces $s \in (u, v)$ and $t \in (v, w)$ with $f(s) = f(t) = c$, contradicting injectivity. $\square$

The proof is the intermediate value theorem used three times: if f were neither increasing nor decreasing, some triple $u < v < w$ would have $f(v)$ outside the interval with endpoints $f(u)$, $f(w)$, and 4.23 would then produce a repeated value on one side or the other.⁵⁵

S4.13 Theorem. *(Inverse functions without compactness.) Let f be continuous and strictly increasing on an interval I . Then $J = f(I)$ is an interval, $f : I \rightarrow J$ is a bijection, and $f^{-1} : J \rightarrow I$ is continuous and strictly increasing.*

SOURCE: ECNU 定理 4.8 (bk. pp. 73–74); USTC 定理 3; Tao I, Prop. 9.8.3.

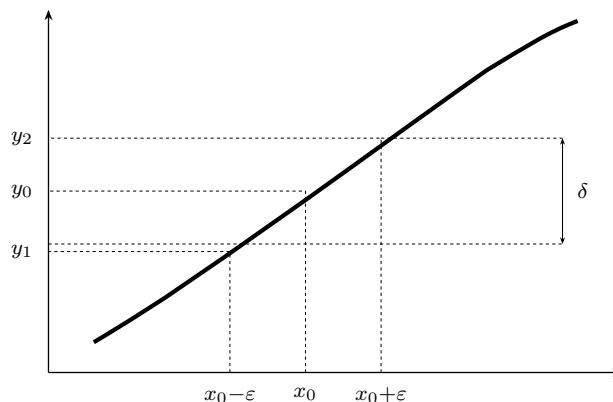
The δ that monotonicity hands you. Let $y_0 = f(x_0)$ with x_0 interior to I , and let $\varepsilon > 0$ be small enough that $[x_0 - \varepsilon, x_0 + \varepsilon] \subset I$. Put $y_1 = f(x_0 - \varepsilon)$ and $y_2 = f(x_0 + \varepsilon)$; strict increase gives $y_1 < y_0 < y_2$. Take

$$\delta = \min\{y_0 - y_1, y_2 - y_0\} > 0.$$

If $|y - y_0| < \delta$ then $y_1 < y < y_2$, and applying the increasing map f^{-1} gives $x_0 - \varepsilon < f^{-1}(y) < x_0 + \varepsilon$. Endpoints of I are handled one-sidedly.

$\square$

⁵⁵ The converse is trivial, so this says: for continuous functions on an interval, injectivity and strict monotonicity are the same hypothesis. Rudin's Exercise 4.14 (open maps are monotone) is a relative of this; the equivalence itself he never states.



Theorem S4.13: the ε -interval about x_0 is carried by f onto an interval about y_0 , and δ is simply the smaller of its two halves. Monotonicity is doing all the work — it is what makes the image of an interval an interval, and what lets f^{-1} preserve the trapping. No compactness enters, so I may be open or unbounded.

Two theorems about inverses, neither containing the other

Rudin 4.17 and S4.13 are genuinely different results:

	Rudin 4.17		S4.13
domain	compact space	metric	interval in $\mathbb{R}$
hypothesis	continuous bijection		continuous, strictly monotone
mechanism	compactness	(4.14, 2.35)	order + IVT
reaches	$\mathbb{R}^k$, function spaces		$(0, \infty)$, $\mathbb{R}$, half-open
misses	$(0, 1) \rightarrow (0, 1)$		anything not ordered

Example 4.21 — the circle-wrap $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi)$, whose inverse is discontinuous — shows 4.17 fails without compactness; but $f(x) = x/(1-x^2)$ on $(-1, 1)$ is a continuous increasing bijection onto $\mathbb{R}$ with continuous inverse, on a domain that is not compact. Order supplies what compactness supplied.

S4.14 Theorem. (*Continuity of the elementary functions.*) For $a > 0$ the function a^x , defined for irrational x as $\sup\{a^r : r \in \mathbb{Q}, r \leq x\}$, is continuous on $\mathbb{R}$ and satisfies $a^\alpha a^\beta = a^{\alpha+\beta}$ and $(a^\alpha)^\beta = a^{\alpha\beta}$. Its inverse $\log_a$ is continuous on $(0, \infty)$ by S4.13, the power $x^\alpha = e^{\alpha \ln x}$ is continuous on $(0, \infty)$, and $\sin, \cos$ are continuous because $|\sin x - \sin y| \leq |x - y|$. Consequently every function built from these by finitely many algebraic operations, compositions, and inversions is continuous on its domain of definition.

SOURCE: ECNU 定理 4.10–4.13 (bk. pp. 78–80); USTC 定理 4 (bk. pp. 45–46).

Proof sketch. Continuity of a^x at 0 follows from $a^{1/n} \rightarrow 1$ (Rudin 3.20(b)) and monotonicity; the identity $a^{x+h} = a^x a^h$ then transports

continuity from 0 to every point, since $|a^{x+h} - a^x| = a^x|a^h - 1|$. The laws $a^\alpha a^\beta = a^{\alpha+\beta}$ and $(a^\alpha)^\beta = a^{\alpha\beta}$ hold for rational exponents by S1.13 and extend to real exponents by density and continuity (Rudin's Exercise 4.4). Continuity of $\log_a$ is S4.13, that of $x^\alpha = e^{\alpha \ln x}$ is composition, and $|\sin x - \sin y| \leq |x - y|$ gives the trigonometric cases. Closure under sums, products, quotients and composition is 4.9 and 4.7. $\square$

The same destination by two routes

Rudin builds these functions in Chapter 8 from power series, and their continuity is then a corollary of the theorem that a power series is continuous inside its disc of convergence (8.1). The Chinese texts build them here, from the supremum definition of a^x that Rudin sets as Exercises 1.6–1.7, and get continuity by hand.

The trade is visible: Rudin's route is shorter and yields differentiability and the addition formulas at the same stroke, but it defers every concrete function by four chapters; the elementary route has functions available immediately but must prove each property separately. Neither is more rigorous. The Chinese texts' order is why their students compute limits in week three and Rudin's readers do not.

NOT IN RUDIN

SUPPLEMENT E — UNIFORM CONTINUITY: THE WORKING TOOLKIT

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Definition 4.18 gives uniform continuity, 4.19 proves that continuity on a compact set implies it, and 4.20 exhibits the standard failures. Everything else about the notion is set as Exercises 4.8–4.13.

The gap. Those exercises contain the working theory — uniform continuity preserves Cauchy sequences, preserves boundedness, permits extension to the closure — and as exercises they are unavailable to a reader who has not done them, and unquotable to one who has. There is also no *criterion*: to show a function is not uniformly continuous, Rudin negates the definition by hand each time.

Where the gap bites. In Chapter 8, where b^x is defined on $\mathbb{Q}$ and must be extended to $\mathbb{R}$ — an appeal to Exercise 4.13; and at 7.16 and beyond, where uniform continuity is used as a known tool.

What this section supplies. The sequential criterion, which turns every non-example into a two-line computation; the gluing

lemma; the extension theorem in its general form; the algebra of what preserves uniform continuity; and the Lebesgue number lemma, which is what Rudin's own proof of 4.19 manufactures without naming.

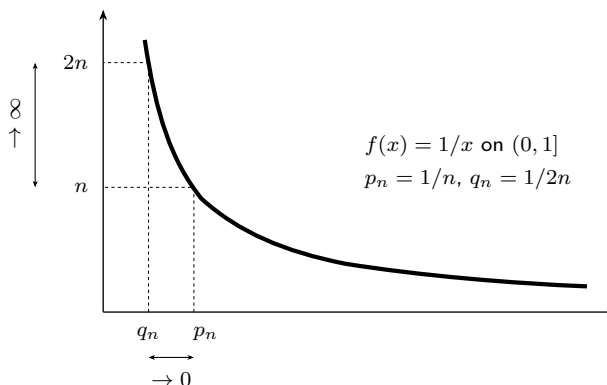
S4.15 Theorem. (*Sequential criterion.*) $f : E \rightarrow Y$ is uniformly continuous on E if and only if for every pair of sequences $(p_n), (q_n)$ in E with $d(p_n, q_n) \rightarrow 0$ one has $d(f(p_n), f(q_n)) \rightarrow 0$.

SOURCE: Tao I, Prop. 9.9.8 (via “equivalent sequences”); ECNU 例 10 (bk. p. 75).

Proof. If f is uniformly continuous and $d(p_n, q_n) \rightarrow 0$, then given $\varepsilon > 0$ take δ from the definition; eventually $d(p_n, q_n) < \delta$, so eventually $d(f(p_n), f(q_n)) < \varepsilon$.

Conversely, suppose f is not uniformly continuous. Then some $\varepsilon > 0$ resists every δ ; taking $\delta = 1/n$ produces points p_n, q_n with $d(p_n, q_n) < 1/n$ but $d(f(p_n), f(q_n)) \geq \varepsilon$. This pair of sequences violates the stated condition. $\square$

The value is in the negation: to prove a function *not* uniformly continuous, exhibit one pair of sequences that close up while their images do not.⁵⁶



The criterion applied to $1/x$: the two arguments close up ($p_n - q_n \rightarrow 0$) while their values pull apart (n versus $2n$). One pair of sequences settles it — no ε - δ negation needed. For x^2 on $\mathbb{R}$ take $p_n = n, q_n = n + 1/n$: the gap tends to 0 and the images differ by $2 + 1/n^2$.

S4.16 Theorem. (*Gluing.*) If $a < c < b$ and f is uniformly continuous on $[a, c]$ and on $[c, b]$, then f is uniformly continuous on $[a, b]$. The same holds for two intervals sharing one endpoint, bounded or not.

SOURCE: ECNU 例 12 (bk. p. 76).

Proof. Given $\varepsilon > 0$, take δ_1 and δ_2 for $\varepsilon/2$ on $[a, c]$ and on $[c, b]$, and put $\delta = \min\{\delta_1, \delta_2\}$. Let $|x - y| < \delta$. If x, y lie in the same piece there is nothing to prove. Otherwise $x \leq c \leq y$, so $|x - c| < \delta$ and $|c - y| < \delta$, and

$$|f(x) - f(y)| \leq |f(x) - f(c)| + |f(c) - f(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

⁵⁶ Compare Rudin's own method at 4.20: he argues directly from the ε - δ negation each time. The criterion packages that argument once. Note the parallel it draws: continuity preserves convergent sequences (4.2), uniform continuity preserves pairs that close up.

□

WHY GLUING NEEDS AN ARGUMENT AT ALL, AND WHAT IT BUYS

Uniform continuity is not a local property, so the union of two sets on which f is uniformly continuous is not automatically another. The proof works because the two pieces meet: given ε , take δ_1, δ_2 for $\varepsilon/2$ on the two pieces and set $\delta = \min\{\delta_1, \delta_2\}$. If $|x - y| < \delta$ with $x \leq c \leq y$, then $|x - c| < \delta$ and $|c - y| < \delta$, so

$$|f(x) - f(y)| \leq |f(x) - f(c)| + |f(c) - f(y)| < \varepsilon.$$

The shared point c is the whole mechanism, and it is why two intervals with a *gap* between them cannot be glued.

The payoff is a standard trick: to prove f uniformly continuous on $[0, \infty)$, prove it on the compact $[0, 1]$ by 4.19 and on $[1, \infty)$ by a direct estimate. For $f(x) = \sqrt{x}$ the second is $|\sqrt{x} - \sqrt{y}| = |x - y|/(\sqrt{x} + \sqrt{y}) \leq \frac{1}{2}|x - y|$. Rudin's Exercise 4.8 asks for the boundedness half of this story and never supplies the gluing tool.

S4.17 Theorem. (*Extension to the closure.*) Let $f : E \rightarrow Y$ be uniformly continuous with Y complete. Then $\lim_{x \rightarrow p} f(x)$ exists at every $p \in \overline{E}$, and f extends uniquely to a uniformly continuous function on $\overline{E}$.

SOURCE: Tao I, Cor. 9.9.14 and Prop. 9.9.12. Rudin poses the special case as Exercise 4.13.

Proof. Let $p \in \overline{E}$ and let (p_n) be a sequence in E with $p_n \rightarrow p$. It is Cauchy, and a uniformly continuous map sends Cauchy sequences to Cauchy sequences: given ε , take δ from uniform continuity and then N with $d(p_n, p_m) < \delta$ for $n, m \geq N$. Completeness of Y makes $(f(p_n))$ converge. Interleaving two such sequences shows the limit does not depend on the choice, so $\lim_{x \rightarrow p} f(x)$ exists by 4.2.

Defining $\tilde{f}(p)$ to be that limit extends f to $\overline{E}$; the extension is uniformly continuous because the defining inequality passes to limits, and it is unique because two continuous maps agreeing on the dense set E agree on $\overline{E}$ (Rudin's Exercise 4.4). □

HOW THE PROOF FLOWS

1. **Choose the method.** A *value* has to be produced at a point where f is not defined, so something must manufacture it. The only manufacturing device available is completeness of the target, and the only way to reach it is through Cauchy sequences.
2. **Convert the hypothesis into a Cauchy statement.** Uniform continuity says one δ works everywhere, which is exactly what lets a Cauchy sequence in the domain be pushed to a Cauchy sequence in the target. Ordinary continuity cannot do this — the δ would move with the point.

3. **Spend completeness.** Cauchy becomes convergent. This is the only step that uses anything about Y .
4. **Remove the dependence on the sequence.** Interleaving two approximating sequences is the standard device; it converts “every sequence gives a limit” into “the limit exists,” which is 4.2.
5. **Audit.** Uniform continuity is spent once, in step 2; completeness once, in step 3; density of E in $\overline{E}$ once, to know the approximating sequences exist at all. Drop uniformity and the theorem fails — $\sin(1/x)$ on $(0, 1)$ is continuous, sends a Cauchy sequence to a divergent one, and extends to no value at 0.

This is the theorem Rudin’s Exercise 4.13 is a special case of, and the reason uniform continuity matters at all in Chapter 8: it is what lets a function defined on $\mathbb{Q}$ — like b^x for rational x in Exercise 1.6 — become a function on $\mathbb{R}$.⁵⁷

S4.18 Theorem. (What preserves uniform continuity.) If f, g are uniformly continuous on E then so are $f+g$, $f-g$, cf , $\max(f, g)$, $\min(f, g)$, and $g \circ f$ where defined. The product fg need not be, but is if f and g are both bounded.

SOURCE: Tao I, Ex. 9.9.6; Tao II, Ex. 2.3.4–2.3.6.

Proof. For $f+g$ take δ to be the smaller of the two δ ’s for $\varepsilon/2$; for cf use $\varepsilon/|c|$; for $g \circ f$ feed the δ of g into f as its ε . The identity $\max(f, g) = \frac{1}{2}(f+g+|f-g|)$ and $||u|-|v|| \leq |u-v|$ handle max and min.

For products with both factors bounded by M ,

$$\begin{aligned} |f(p)g(p) - f(q)g(q)| &\leq |f(p)||g(p) - g(q)| + |g(q)||f(p) - f(q)| \\ &\leq M(|g(p) - g(q)| + |f(p) - f(q)|), \end{aligned}$$

so δ for $\varepsilon/2M$ in each factor works. Without boundedness the conclusion fails: $f(x) = g(x) = x$ are uniformly continuous on $\mathbb{R}^1$ and $fg = x^2$ is not. $\square$

The counterexample is the smallest one available: $f(x) = g(x) = x$ is uniformly continuous on $\mathbb{R}$, and $fg = x^2$ is not — the same function Rudin uses at 4.20 for a different purpose.⁵⁸

S4.19 Theorem. (Lebesgue number.) Let K be a compact metric space and $\{U_\alpha\}$ an open cover of K . Then there is a $\delta > 0$ — a Lebesgue number of the cover — such that every ball $B(x, \delta)$ with $x \in K$ lies inside a single U_α .

SOURCE: Yu Pin, Prop. 66 (p. 112) and the metric version p. 114.

Proof. Suppose no such δ exists. Then for each n there is $x_n \in K$ with $B(x_n, 1/n)$ contained in no single U_α . Compactness gives a subsequence

⁵⁷ Chain of reasoning: uniformly continuous maps send Cauchy sequences to Cauchy sequences (Rudin’s Exercise 4.11), the target is complete, so the images converge; S4.1 then upgrades “every sequence works” to “the limit exists.” Plain continuity is not enough: $1/x$ on $(0, 1)$ is continuous, sends the Cauchy sequence $(1/n)$ to an unbounded one, and does not extend to 0.

⁵⁸ The boundedness repair is the usual splitting: $|fg(p) - fg(q)| \leq |f(p)||g(p) - g(q)| + |g(q)||f(p) - f(q)|$, and if both factors are bounded by M each term is small. Compare 3.3(c), where exactly this decomposition proves the product rule for limits — the strips-and-corner picture there is the same algebra.

$x_{n_k} \rightarrow x$, and x lies in some U_β , which contains a ball $B(x, r)$. For k large enough that $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, the triangle inequality gives $B(x_{n_k}, 1/n_k) \subset B(x, r) \subset U_\beta$ — contradicting the choice of x_{n_k} . $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** The claim is that a *single* scale works everywhere, and no candidate scale is in sight. So assume none exists and let the failure build a sequence — the standard move when a uniform quantity is wanted and only pointwise data is given.
2. **Let the counterexamples index themselves.** Failure at scale $1/n$ produces a point x_n . The sequence is free: it is the negation of the conclusion, written out.
3. **Spend compactness.** A convergent subsequence appears, and its limit lies in some cover element — which finally supplies a ball of a definite radius r , the first fixed scale in the argument.
4. **Collide the two scales.** For large k the tiny ball around x_{n_k} sits inside $B(x, r)$, hence inside one U_β — contradicting how x_{n_k} was chosen. Note both smallnesses are needed: $d(x_{n_k}, x)$ small and $1/n_k$ small.
5. **Audit.** Compactness is spent once, in step 3; openness of the cover once, also in step 3, to get r . The lemma is what Rudin's own 4.19 constructs by hand — his halving of $\phi(p)$ and taking the minimum over a finite subcover produces exactly this δ .

THE LEMMA HIDING INSIDE RUDIN'S PROOF OF 4.19

Rudin's uniform continuity proof does not name this, but it manufactures it: he takes the balls $J(p)$ of radius $\frac{1}{2}\phi(p)$, extracts a finite subcover, and sets $\delta = \frac{1}{2} \min \phi(p_i)$. That δ is a Lebesgue number of the cover $\{J(p)\}$, and the halving trick is exactly the device that produces one.

Isolate the lemma and 4.19 becomes three lines. Given $\varepsilon > 0$, cover K by the sets $U_x = f^{-1}(B(f(x), \varepsilon/2))$, open by 4.8. Let δ be a Lebesgue number. If $d(p, q) < \delta$ then both lie in one U_x , so $f(p)$ and $f(q)$ are both within $\varepsilon/2$ of $f(x)$, hence within ε of each other. The gain is not brevity but reuse: the same lemma proves that compact and sequentially compact agree, and it is the standard tool whenever a *single* scale must be extracted from a cover. Yu Pin runs Heine–Borel itself off it. Keep it as the answer to “what does compactness actually hand me?” — a uniform scale.

SUPPLEMENT F — FINER STRUCTURE: DISCONTINUITIES AND COMPONENTS

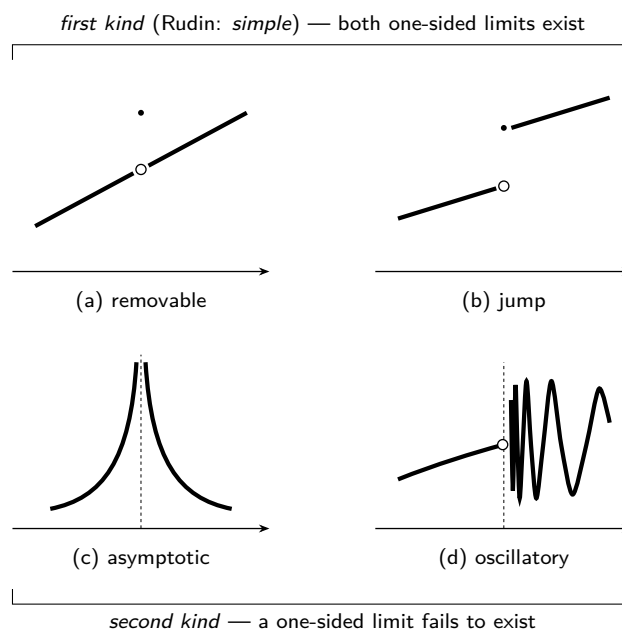
S4.20 Definition. (Four kinds of discontinuity.) Let f be discontinuous at x . The discontinuity is

- (a) **removable** if $f(x+)$ and $f(x-)$ exist and are equal (necessarily $\neq f(x)$);
- (b) **a jump** if both exist and differ;
- (c) **asymptotic** if a one-sided limit fails because $|f|$ is unbounded near x on that side;
- (d) **oscillatory** if a one-sided limit fails while f stays bounded there.

SOURCE: Tao I, Remark 9.5.4; USTC 可去间断点/跳跃点, bk. pp. 42–43.

Rudin's Definition 4.26 stops at two boxes: (a) and (b) together are his *simple* discontinuities, and (c) and (d) together are his *second kind*.⁵⁹

59 The refinement earns its keep twice. Removable discontinuities are exactly the ones a change of a single value repairs, so they are not really properties of the function but of its bookkeeping. And the asymptotic/oscillatory split is the difference between $1/x$ and $\sin(1/x)$ at 0 — two failures that behave nothing alike, which Rudin's 4.27(d) illustrates without naming.



Rudin's two boxes opened into four. His 4.27(c) is case (b) and his 4.27(d) is case (d); cases (a) and (c) he never names. The classification is per side — a function may be removable from the left and asymptotic from the right — so these are the four pure types, not a partition of all discontinuities.

S4.21 Definition. $E \subset X$ is *path-connected* if for all $p, q \in E$ there is a continuous $\gamma : [0, 1] \rightarrow E$ with $\gamma(0) = p$, $\gamma(1) = q$. Every path-connected set is connected; the converse is false.

SOURCE: Tao II, Ex. 2.4.7 and §2.4.

The counterexample is Rudin's own $\sin(1/x)$

That path-connected implies connected is Rudin 4.22 applied to γ : if $E = A \cup B$ were a separation, the sets $\gamma^{-1}(A)$ and $\gamma^{-1}(B)$ would separate $[0, 1]$, which 2.47 forbids.

For the converse, take the closure of

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\},$$

which adds the segment $\{0\} \times [-1, 1]$. It is connected — S is connected as a continuous image of $(0, 1]$, and adding limit points of a connected set preserves connectedness — but no path joins a point of the segment to a point of S : a path would have to traverse infinitely much oscillation in finite time, and one shows its first coordinate cannot become positive. So Rudin's favourite pathological function, introduced at 4.27(d) purely as a discontinuity, turns out to also separate two notions of connectedness he never distinguishes.

S4.22 Theorem. (*Components.*) Define $p \sim q$ if some connected subset of X contains both. This is an equivalence relation; its classes, the connected components of X , are connected and closed, and they partition X .

SOURCE: Tao II, Ex. 2.4.6, 2.4.8, 2.4.9.

Proof. Reflexivity holds because $\{p\}$ is connected, and symmetry is trivial. For transitivity, if A is connected containing p, q and B is connected containing q, r , then $A \cup B$ is connected — a separation of $A \cup B$ would separate one of A, B , since q lies in both — and it contains p and r .

The class of p is the union of all connected sets containing p ; that union is connected by the same argument, so each class is connected and is maximal among connected sets meeting it. Finally the closure of a connected set is connected, so a class equals its own closure and is therefore closed. $\square$

Transitivity holds because a union of connected sets with a point in common is connected, and closedness because the closure of a connected set is connected.⁶⁰

S4.23 Theorem. (*Convexity, five ways.*) For f on an interval I the following are equivalent: (i) $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for $\lambda \in (0, 1)$; (ii) for $x_1 < x_2 < x_3$ in I ,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2};$$

(iii) for each fixed a , the slope $x \mapsto (f(x) - f(a))/(x - a)$ is increasing; (iv) the 3×3 determinant with rows $(1, 1, 1)$, (x_1, x_2, x_3) , $(f(x_1), f(x_2), f(x_3))$ is ≥ 0 ; (v) the epigraph $\{(x, y) : y \geq f(x)\}$ is a convex subset of $\mathbb{R}^2$.

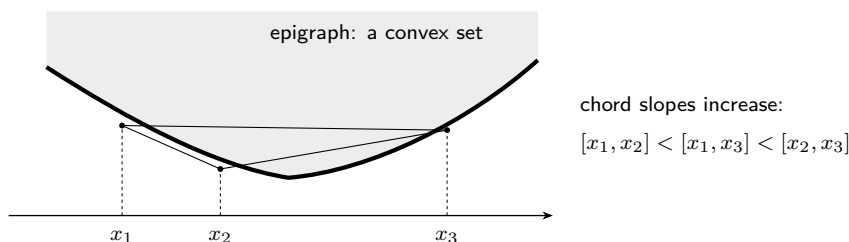
SOURCE: Yu Pin, Thm 106 (pp. 175–177); ECNU 凸函数 §6.5 gives (i)–(iii).

Proof of (i) $\Leftrightarrow$ (ii). Given $x_1 < x_2 < x_3$, write $x_2 = \lambda x_1 + (1 - \lambda)x_3$ with $\lambda = (x_3 - x_2)/(x_3 - x_1) \in (0, 1)$. Then (i) says $f(x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_3)$, and subtracting $f(x_1)$ from both sides and dividing by $x_2 - x_1 = (1 - \lambda)(x_3 - x_1) > 0$ turns this into the first inequality of (ii);

⁶⁰ Rudin never decomposes a space this way, though 2.47 is the statement that the components of a subset of $\mathbb{R}$ are intervals, and Exercise 2.29 — every open set in $\mathbb{R}$ is a countable union of disjoint segments — is the component decomposition of an open set. Naming the notion makes both results one result.

subtracting instead from $f(x_3)$ and dividing by $x_3 - x_2$ gives the second. Each step is reversible, so (ii) implies (i).

Statement (iii) is (ii) with x_1 held fixed, (iv) is (ii) cleared of denominators, and (v) says that the chord joining two points of the graph lies in the region above the graph — which is (i) verbatim. $\square$



The chord slopes over $[x_1, x_2]$, $[x_1, x_3]$, $[x_2, x_3]$ increase from left to right — statement (ii), which is (i) rewritten with $\lambda = (x_3 - x_2)/(x_3 - x_1)$. Statement (v) says the same thing without picking any points at all: everything on or above the graph is a convex set, which is what “the chord lies above the graph” means.

Why five formulations instead of one

Rudin’s Exercises 4.23–4.24 give the definition and ask for continuity on the open interval plus the midpoint characterisation. The equivalences above are what make those exercises routine, and each is the right tool for a different job. Form (ii) is the one that proves continuity: the slopes from a fixed interior point are bounded on both sides, so f is Lipschitz on any compact subinterval of I° — which incidentally shows why Rudin’s exercise is stated on the *open* interval, since an endpoint can carry a jump upward and stay convex.

Form (v) is the conceptual one: it converts a statement about a function into a statement about a *set*, so that convexity of functions and convexity of sets stop being two ideas. Form (iv) is the computational one, and form (iii) is what one differentiates in Chapter 5 to get “ f' increasing” (Theorem 5.11’s setting).

CONTINUITY ALONG LINES. SEPARATELY AND JOINTLY

Three conditions on $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at a point, each strictly weaker than the next:

- (a) continuous in each variable separately (two lines);
- (b) continuous along every straight line through the point;
- (c) jointly continuous.

The function $xy/(x^2 + y^2)$, with value 0 at the origin, separates (a) from (b): it vanishes on both axes but equals $m/(1 + m^2)$ along $y = mx$, so it is separately continuous and not even constant along lines. Rudin’s Exercise 4.7 does the harder separation of (b) from (c): $xy^2/(x^2 + y^4)$ is continuous along every line yet approaches $\frac{1}{2}$ along the parabola $x = y^2$.

Keeping the hierarchy straight matters in Chapter 9, where partial derivatives are a statement of type (a) — which is why 9.21 needs continuity of the partials, not merely their existence.

WHAT THE SUPPLEMENTS CHANGED

Take away four things.

Criteria produce limits; Rudin's theorems transport them. S4.1 and S4.17 are the existence tools, and both need completeness of the target — the same property that made Chapter 1 necessary.

o , O , $\sim$ are the working language of limits, and they are legal in factors, never in summands (S4.7). Rudin does without them because he never computes; you will not have that luxury.

Compactness is not the only source of good behaviour.

Order does some of the same work: monotone functions have continuous inverses on any interval (S4.13), where Rudin's 4.17 needs compactness. Neither theorem contains the other.

Rudin's classifications are coarse on purpose. Two kinds of discontinuity become four (S4.20), connectedness sprouts a path-connected refinement (S4.21), and continuity at a point becomes a *number*, $\omega_f(p)$, which turns “where is f continuous?” into a question with a theorem for an answer (S4.5).

NOT IN RUDIN

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S TWENTY-SIX

Rudin's own twenty-six exercises, which build *his* machinery — the uniform-continuity chain, the ρ_E toolbox, the counterexample gallery — end the chapter proper. These twenty-five come from the four comparison texts and fill the other half of the subject.

Group I (1–6): limits and asymptotics. Computational problems in the $o/\sim$ language of Supplement B, plus three rigidity problems — a periodic function that dies at infinity is dead everywhere, and its relatives. Nothing in Rudin's set resembles these.

Group II (7–11): monotonicity, inverses, functional equations. Where order replaces compactness. Exercise 9 (Cauchy's functional equation) and 10 are the standard route from “continuous at one point” to “continuous, and linear.”

61 Fix x and push it to infinity by its own period: $f(x) = f(x + nT)$ for every n , and the right side tends to 0. The whole content is that periodicity lets one evaluate a function arbitrarily far out without moving it. Compare Rudin's Exercise 4.4: there density does the transporting, here periodicity does.

62 Iterate the hypothesis backwards: $f(x) = f(x^{1/2}) = f(x^{1/4}) = \dots = f(x^{1/2^n})$, and $x^{1/2^n} \rightarrow 1$ for every $x > 0$. The two limit hypotheses are there only to handle x where the iteration escapes. This is the shape of every functional-equation rigidity argument: find the orbit, find its limit point, evaluate there.

63 The continuous Stolz–Cesàro theorem. Write $f(x) = f(x_0) + \sum_{k < n} [f(x_0 + k + 1) - f(x_0 + k)]$ where $x = x_0 + n$ with x_0 in a fixed unit window, then average: the increments are eventually near A , and the finitely many early ones are diluted by $1/x$. It is 3.20's averaging idea run on a continuous variable.

64 Two sequences settle it: $x_n = 2n\pi$ gives $f = 0$, while $x_n = 2n\pi + \pi/2$ gives $f \rightarrow +\infty$. The point of Supplement B's S4.8 — “unbounded” is a statement about the range, “tends to infinity” is a statement about the tail, and only the second forbids returning.

65 Answers $\frac{1}{2}$ and 1. The explanation is the content of the pitfall at S4.7: each equivalence is accurate to first order, the subtraction annihilates exactly the first order, and what survives lives at third order where neither equivalence says anything. Factor first —
 $\tan x - \sin x = \sin x(1 - \cos x)/\cos x$ —
 and every factor may then be replaced.

Group III (12–16): what the intermediate value theorem really gives. The discrete mean value property, the universal chord theorem, and a contraction mapping five chapters before Rudin needs one (9.23).

Group IV (17–21): uniform continuity as a working tool. Prove the criteria of Supplement E, then use them. Compare with Rudin's 4.8–4.13, which prove the same facts one at a time by hand.

Group V (22–25): structure. Oscillation, prescribed discontinuity sets, path-connectedness, and one spectacular payoff of “continuous plus dense determines everything” — a proof of the Cayley–Hamilton theorem that uses no algebra beyond the diagonalisable case.

Marked ★: harder, or requiring a result from a later chapter.

S4.1. Let f be periodic on $\mathbb{R}^1$ with $\lim_{x \rightarrow +\infty} f(x) = 0$. Prove that f is identically zero. No continuity is assumed.⁶¹

SOURCE: ECNU 习题 3.3 #7; USTC Ch. 1 suppl. #13.

S4.2. Suppose f is defined on $(0, +\infty)$, satisfies $f(x^2) = f(x)$ for all x , and $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow +\infty} f(x) = f(1)$. Prove that f is constant.⁶²

SOURCE: ECNU 总练习题 3 #13.

S4.3. ★ Let f be defined on $(a, +\infty)$ and bounded on every finite interval (a, b) , and suppose $\lim_{x \rightarrow +\infty} [f(x+1) - f(x)] = A$. Prove that $\lim_{x \rightarrow +\infty} f(x)/x = A$.⁶³

SOURCE: ECNU 总练习题 3 #14; Yu Pin, HW5 A9* (p. 127), where it is attributed to Cauchy rather than to Stolz.

S4.4. Show that $f(x) = x \sin x$ is unbounded on $(0, +\infty)$ but is *not* an infinitely large quantity as $x \rightarrow +\infty$; that is, $\lim_{x \rightarrow +\infty} f(x) \neq \infty$.⁶⁴

SOURCE: ECNU bk. p. 60; USTC Ex. 1.2.12.

S4.5. Using equivalent infinitesimals, evaluate $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x}$ and $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{1 - \cos x}$. Then explain precisely why substituting $\tan x \sim x$ and $\sin x \sim x$ into the numerator of the first gives the wrong answer 0.⁶⁵

SOURCE: ECNU 例 2 and note, bk. p. 59.

- S4.6.** Find all asymptotes of the curve $y = x^3/(x^2 + 2x - 3)$, and prove that your oblique asymptote satisfies the definition (the distance from $(x, f(x))$ to the line tends to 0).⁶⁶

SOURCE: ECNU 例 6, bk. pp. 61–62.

- S4.7.** Let f be continuous and one-to-one on an interval I . Prove that f is strictly monotone on I . Deduce that a continuous injection of $[a, b]$ into $\mathbb{R}^1$ has a continuous inverse, without invoking compactness.⁶⁷

SOURCE: Tao I, Ex. 9.8.3; USTC 定理 3; Yu Pin, Thm 54.

- S4.8.** Prove that there exists a function $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ that is discontinuous at every point while $|f|$ is continuous at every point.⁶⁸

SOURCE: USTC Ex. 2.1.5.

- S4.9.** Suppose $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ satisfies $f(x + y) = f(x) + f(y)$ for all x, y , and is continuous at the single point 0. Prove that f is continuous on all of $\mathbb{R}^1$ and that $f(x) = f(1)x$.⁶⁹

SOURCE: ECNU 总练习题 4 #9.

- S4.10.** Let f be defined on $\mathbb{R}^1$, continuous at the two points 0 and 1 only (as far as is assumed), and satisfy $f(x^2) = f(x)$ for all x . Prove that f is constant.⁷⁰

SOURCE: ECNU 总练习题 4 #10.

- S4.11.** Determine, with proof, which of the following are possible for a continuous real function: domain $[0, 1]$ with range $(0, \infty)$; domain $[0, 1]$ with range $(0, 1)$; domain $[0, 1]$ with range $[0, 1] \cup [2, 4]$; domain $(0, 1)$ with range $[2, \infty)$.⁷¹

SOURCE: USTC Ex. 2.2.10.

- S4.12.** Let f be continuous on $[a, b]$ and let $x_1, \dots, x_n \in [a, b]$. Prove that there is a $\xi \in [a, b]$ with

$$f(\xi) = \frac{f(x_1) + \dots + f(x_n)}{n}.$$

More generally, prove the same for any weighted average $\sum q_i f(x_i)$ with $q_i > 0$ and $\sum q_i = 1$.⁷²

SOURCE: ECNU 习题 4.2 #19; USTC Ex. 2.2.7.

- S4.13.** (*Universal chord theorem.*) Let f be continuous on $[0, 1]$ with $f(0) = f(1)$. Prove that for every positive integer n there is a $\xi \in [0, 1 - 1/n]$ with

$$f\left(\xi + \frac{1}{n}\right) = f(\xi).$$

66 Long division is the whole computation; the definition check is the observation that the distance to $y = ax + b$ is $|f(x) - ax - b|/\sqrt{1 + a^2}$, a fixed multiple of the vertical gap. Worth doing once to see that “vertical gap” and “perpendicular distance” differ only by that constant — which is why the limits defining a and b may be taken vertically.

67 Suppose not: then some $u < v < w$ has $f(v)$ outside the interval with endpoints $f(u)$ and $f(w)$. Apply the intermediate value theorem on $[u, v]$ and on $[v, w]$ to produce two points with the same value. The deduction is S4.13. Contrast Rudin 4.17, which gets the same conclusion from compactness and works in any metric space — neither theorem contains the other.

68 Modify Dirichlet: $f = 1$ on the rationals, -1 elsewhere. Then $|f| \equiv 1$. The lesson is that $|\cdot|$ destroys information — the implication “ f continuous $\Rightarrow |f|$ continuous” has no converse, and every proof that passes to absolute values gives something away.

69 Cauchy’s functional equation. Continuity everywhere follows because $f(x + h) - f(x) = f(h)$ — the equation makes the increment independent of position, so continuity at one point is continuity everywhere. Then the equation forces $f(q) = f(1)q$ for rational q by induction, and Rudin’s Exercise 4.4 (agreement on a dense set) finishes it. Without some regularity the conclusion is false, but the counterexamples need a Hamel basis and hence the axiom of choice.

70 Sharper than S4.2: continuity at exactly two points suffices. Note f is even, so it is enough to work on $x \geq 0$; for $x > 0$ the iteration $x^{1/2^n} \rightarrow 1$ uses continuity at 1, and $f(0) = \lim_{x \rightarrow 0} f(x)$ uses continuity at 0. Observe which points the iteration funnels into — those are exactly the points where continuity is required.

71 Four probes of the same two theorems. The first three are impossible — 4.15 makes the image compact, hence closed and bounded, and 4.22 makes it connected, hence an interval. The fourth is possible; construct one. A good check on whether you know what compactness and connectedness each contribute.

72 The average lies between $\min_i f(x_i)$ and $\max_i f(x_i)$, both of which are attained values of f ; now apply the intermediate value theorem between the two points where they are attained. This is the discrete ancestor of the mean value theorem for integrals (6.x) and of the notion of a barycentre.

73 Put $F(x) = f(x + 1/n) - f(x)$ on $[0, 1 - 1/n]$ and sum:
 $F(0) + F(1/n) + \cdots + F((n-1)/n) = f(1) - f(0) = 0$. A finite list of reals summing to zero either is all zeros or contains both signs, so F has a zero by 4.23. The telescoping sum is the whole idea, and it is why only reciprocal lengths work.

74 Existence and uniqueness from 4.23 plus strict increase of the left side on $(0, \infty)$. For the limit, note $x_n \in (\frac{1}{2}, 1)$ and that the sum telescopes:
 $x(1 - x^n)/(1 - x) = 1$. Show (x_n) is decreasing and bounded below, then pass to the limit in the closed form to get $\frac{1}{2}$. A clean synthesis of Chapters 3 and 4.

75 Uniqueness is immediate from the inequality. For existence, do not use 4.23 — use the iteration:
 $|x_{n+1} - x_n| \leq k^{n-1}|x_2 - x_1|$ makes (x_n) Cauchy by comparison with a geometric series (3.26), and 3.11 gives the limit, which is a fixed point by continuity. This is Rudin's 9.23 in one dimension, and the proof is identical — worth doing now so that 9.23 is recognition rather than novelty. Compare Rudin's Exercise 4.14, which gets a fixed point from the intermediate value theorem alone but no iteration.

Show by example that the conclusion can fail for a chord length that is not of the form $1/n$.⁷³

SOURCE: ECNU 总练习题 4 #8; USTC Ch. 2 suppl. #5.

S4.14. Prove that for every positive integer n the equation

$$x^n + x^{n-1} + \cdots + x = 1$$

has exactly one positive root x_n ; prove that (x_n) converges, and find its limit.⁷⁴

SOURCE: USTC Ch. 2 suppl. #9.

S4.15. (Contraction mapping on an interval.) Let f map $[a, b]$ into $[a, b]$ and suppose there is a constant k with $0 < k < 1$ such that

$$|f(x) - f(y)| \leq k|x - y| \quad (x, y \in [a, b]).$$

Prove that f has a unique fixed point x_0 , and that for every starting value $x_1 \in [a, b]$ the sequence $x_{n+1} = f(x_n)$ converges to x_0 .⁷⁵

SOURCE: USTC Ch. 2 suppl. #8.

S4.16. $\star$ (Croft.) Let f be continuous on $\mathbb{R}^1$ and suppose that for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} f(n\delta) = 0.$$

Prove that $\lim_{x \rightarrow +\infty} f(x) = 0$.⁷⁶

SOURCE: Yu Pin, HW5 problem G (p. 129).

S4.17. Prove the sequential criterion S4.15: f is uniformly continuous on E if and only if $d(f(p_n), f(q_n)) \rightarrow 0$ for every pair of sequences in E with $d(p_n, q_n) \rightarrow 0$. Then use it to show that $\sin(1/x)$ is not uniformly continuous on $(0, 1)$ and that x^2 is not uniformly continuous on $\mathbb{R}^1$.⁷⁷

SOURCE: Tao I, Prop. 9.9.8; ECNU 例 10–11.

S4.18. Prove the gluing lemma S4.16, and use it with Theorem 4.19 to show that $\sqrt{x}$ is uniformly continuous on $[0, +\infty)$. Show likewise that $\cos \sqrt{x}$ is uniformly continuous there.⁷⁸

SOURCE: ECNU 例 12 and 习题 4.2 #12, #20.

S4.19. Let f be continuous on $[a, +\infty)$ and suppose $\lim_{x \rightarrow +\infty} f(x)$ exists and is finite. Prove that f is bounded on $[a, +\infty)$ and uniformly continuous there. Must f attain a maximum? ⁷⁹

SOURCE: ECNU 习题 4.2 #6, #16.

- S4.20.** Prove that if f and g are uniformly continuous and *bounded* on E , then fg is uniformly continuous on E ; and show by example that boundedness cannot be dropped. Which of $f+g$, $\max(f, g)$, $g \circ f$ are uniformly continuous without extra hypotheses?⁸⁰

SOURCE: *Tao II*, Ex. 2.3.4–2.3.6.

- S4.21.** Prove S4.17: a uniformly continuous $f : E \rightarrow \mathbb{R}^k$ has a limit at every point of $\overline{E}$, and therefore extends uniquely to a uniformly continuous function on $\overline{E}$. Show that $\sin(1/x)$ on $(0, 1)$ has no such extension, and identify which hypothesis it violates.⁸¹

SOURCE: *Tao I*, Cor. 9.9.14; generalises Rudin's Exercise 4.13.

- S4.22.** Define the oscillation $\omega_f(p)$ as in S4.4. Prove that f is continuous at p if and only if $\omega_f(p) = 0$, that $\{x : \omega_f(x) \geq \varepsilon\}$ is closed for each $\varepsilon > 0$, and hence that the set of discontinuities of *any* f is a countable union of closed sets. Deduce, using Exercise 3.22, that no function is continuous at every rational and discontinuous at every irrational.⁸²

SOURCE: *Yu Pin*, Lemmas 131–132; *Tao I*, Ex. 9.3.4.

- S4.23.** Let $A \subset \mathbb{R}^1$ be countable. Construct a monotonically increasing $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ whose set of discontinuities is exactly A .⁸³

SOURCE: *Yu Pin*, HW4 problem E (p. 110); cf. *Tao I*, Ex. 9.8.5.

- S4.24.** ★ (*Cayley–Hamilton by continuity.*) For A an $n \times n$ complex matrix let $P_A(X) = \det(XI - A)$. Prove: (a) if A is diagonalisable then $P_A(A) = 0$; (b) the map $A \mapsto P_A(A)$ is continuous on the space of matrices with the norm $\|\cdot\|$; (c) every A is a limit of diagonalisable matrices; (d) therefore $P_A(A) = 0$ for every A .⁸⁴

SOURCE: *Yu Pin*, winter problem set U (pp. 331–332).

- S4.25.** Prove that a path-connected set is connected, and that the closure of

$$S = \{ (x, \sin(1/x)) : 0 < x \leq 1 \}$$

is connected but not path-connected.⁸⁵

SOURCE: *Tao II*, Ex. 2.4.7–2.4.9.

76 Surprisingly hard, and the tool is Baire (Rudin's Exercise 3.22). Suppose the conclusion fails; build closed sets $F_N = \{\delta \geq 1 : |f(n\delta)| \leq \varepsilon \text{ for all } n \geq N\}$, whose union is $[1, \infty)$, so one has interior; then the dilates $\{n\delta : \delta \in I, n \geq N\}$ of an interval I eventually cover a half-line. The moral: a hypothesis quantified over every δ is a category hypothesis in disguise.

77 The forward direction is a substitution. For the converse, negate uniform continuity: some ε has, for each $\delta = 1/n$, a pair p_n, q_n within $1/n$ whose images are ε apart — the failing pair builds itself. For the examples take $1/(2n\pi)$ against $1/(2n\pi + \pi/2)$, and n against $n + 1/n$.

78 Split at 1: on $[0, 1]$ use compactness, on $[1, \infty)$ use $|\sqrt{x} - \sqrt{y}| = |x - y|/(\sqrt{x} + \sqrt{y}) \leq \frac{1}{2}|x - y|$. Note that $\sqrt{x}$ is uniformly continuous although its derivative is unbounded near 0 — so “bounded derivative” is sufficient for uniform continuity but far from necessary.

79 Both halves are the gluing pattern: the limit hypothesis makes f nearly constant past some M , and $[a, M]$ is compact. So a hypothesis at infinity substitutes for compactness of the domain. The last question is no: $f(x) = 1 - 1/x$ on $[1, \infty)$ has supremum 1, never attained.

80 Split the product: $|f(p)g(p) - f(q)g(q)| \leq |f(p)||g(p) - g(q)| + |g(q)||f(p) - f(q)|$ — the identical decomposition as at 3.3(c). For the counterexample $f = g = x$ on $\mathbb{R}^1$. Sum, max and composition all survive with no extra hypotheses.

81 Uniform continuity sends Cauchy sequences to Cauchy sequences (Rudin's Exercise 4.11), completeness converts those to limits, and S4.1 promotes “every sequence converges” to “the limit exists.” Uniqueness is Rudin's Exercise 4.4. The example is merely continuous, not uniformly — by S4.15 with p_n, q_n as in exercise 17.

82 The last part is the converse question to Rudin's Exercise 4.18 (Thomae), and the answer is that the two cases are not symmetric. If such an f existed, $\mathbb{Q}$ would be a countable intersection of dense open sets, and intersecting further with the complements of the individual rationals would exhibit $\emptyset$ as such an intersection — contradicting Baire.

83 The exact converse of 4.30, and Rudin's Remark 4.31 sketches the construction: enumerate $A = \{a_1, a_2, \dots\}$ and put $f(x) = \sum_{a_n < x} 2^{-n}$. Convergence is comparison with a geometric series; the jump at a_n is exactly 2^{-n} ; continuity elsewhere follows because the tail beyond N contributes less than 2^{-N} . Together with 4.30 this characterises the possible discontinuity sets of monotone functions completely — a rare instance of a sharp answer.

84 The best advertisement in this chapter for Rudin's Exercise 4.4. Part (a) is one line in an eigenbasis; (b) holds because the entries of $P_A(A)$ are polynomials in the entries of A ; (c) perturbs the eigenvalues of an upper triangular form to make them distinct. Then a purely algebraic theorem falls out of “two continuous maps agreeing on a dense set are equal” — with the density supplied by linear algebra and the rigidity by Chapter 4.

85 The first part is 4.22 applied to the path. For the second, $\overline{S}$ adds the segment $\{0\} \times [-1, 1]$; connectedness holds because the closure of a connected set is connected, while a path from the segment into S would have to traverse infinitely many oscillations in finite time — make that precise by showing its first coordinate cannot leave 0. Rudin's favourite pathology at 4.27(d) turns out to separate two notions of connectedness he never distinguishes.